

HJ-Gauss

Monte-Carlo Hamilton-Jacobi Reachability

Full Exposition: From TVD-RK Levelsets to Monte Carlo Sampling for BRTs.

Lekan Molu · Amazon IRG

Amazon Robotics · August 18, 2026

with V. Renganathan (Cranfield) and N. Cho (Seoul National University)

BLUF (Bottom Line Up Front) - (1/2)

Grid-resolved Hamilton-Jacobi (HJ) BRTs yield certified safety sets at a $O(M^n)$ memory cost.

-  **Background:** LevelSetPy resolves backward reachable sets/tubes by discretizing the HJ PDE (upwinding + Lax-Friedrichs + TVD-RK) on a grid — certified, but exponential in state dimension n .
-  **Idea:** a Cole-Hopf transformation turns the *viscous* HJ PDE into a **linear heat equation**; Feynman-Kac evaluates it as a **Gaussian expectation**. No grid.

HJ-Gauss replaces the grid with Gaussian Monte-Carlo, dropping the footprint to $O(N \cdot n)$.

BLUF (Bottom Line Up Front) - (2/2)

HJ-Gauss replaces the grid with Gaussian Monte-Carlo, dropping the footprint to $O(N \cdot n)$.

-  **Generality:** a frozen-coefficient **Picard iteration** handles the nonconvex, state-dependent Hamiltonians of reachability.
-  **Rigor:** $O(N^{-1/2})$ concentration, contraction convergence, a linearization residual bound, and a **conservative safety certificate**.
-  **For this room:** the windowed reachable tube is a dynamics-aware **conflict predicate** for MAPF; the value gradient is a **score** for diffusion planners.

This Gaussian kernel is the diffusion kernel, the bridge to learning-based planners.

Roadmap — Parts 0-6

Part	Theme
0	Framing, significance, notation
1	Hamilton-Jacobi PDE theory & viscosity solutions
2	Reachability & safety foundations
3	The grid pipeline: LevelSetPy numerics
4	The wall: curse of dimensionality & prior grid-free work
5	HJ-Gauss core theory: Cole-Hopf $\rightarrow$ heat $\rightarrow$ Feynman-Kac $\rightarrow$ Picard
6	Importance sampling & variance control

 **Continue:** Parts 7-12 on the next slide.



Roadmap — Parts 7-12

Part	Theme
7	Guarantees: concentration, contraction, residual, certificate
8	Experiments from the paper (rockets, Dubins, 45D, 10^5 birds)
9	Application: multi-agent path finding (AMFS)
10	The diffusion connection
11	Dirty Laundry: limits & boundaries
12	Conclusions, future work, references

Notations

Symbol	Meaning
$x \in \Omega \subseteq \mathbb{R}^n$	x : State; Ω : Open set; n = State dimension
$v(t, x)$	Value function; v_t time derivative; $Dv = \nabla_x v$ spatial gradient (co-state)
$H(t; x, p)$	Hamiltonian; p = co-state
$g(x)$	Terminal/target datum (signed distance $\ell(x)$); BUC
$\delta > 0$	Viscosity parameter

Notations

Symbol	Meaning
$\omega^\delta = e^{-cv^\delta}$	Cole-Hopf transformed variable
$c(t; x)$	Frozen coefficient $= \frac{2}{\delta} H^\delta / Dv^\delta ^2$
M, N	Grid points per dimension (M); Monte-Carlo samples per query state (N)
$\mathcal{L}_0, \mathcal{L}$	Target set, backward reachable tube (BRT)

Two independent counts: M evaluation states (arbitrary, grid-free) vs N Gaussian samples drawn *per state*. Total randomness per iteration $= M \times N$.



Who Should Care, and Why

- **Autonomous mobility / MAPF:** a certified, dynamics-aware, disturbance-robust conflict predicate that slots into rolling-horizon planners at fleet scale.
- **Safe RL / policy verification:** computing backward reachable tubes for a closed-loop learned policy is memory-bound on grids; $O(N \cdot n)$ lifts that bound.
- **Differential games / pursuit-evasion:** the adversarial worst-case reachable set is what HJ-Isaacs computes.

The unifying object is a **value function whose sign certifies safety**. This talk is about computing that sign where grids cannot.



Who Should Care, and Why

- **Differential games / pursuit-evasion:** the adversarial worst-case reachable set is what HJ-Isaacs computes.
- **Diffusion / generative planning:** the Cole-Hopf kernel *is* the score-based diffusion kernel; the value gradient is a score.
- **Air-traffic, collision avoidance, multi-robot control:** the classical application domain of HJ reachability.

The unifying object is a **value function whose sign certifies safety**. This talk is about computing that sign where grids cannot.

Part 1

Hamilton-Jacobi PDE Theory Recap

Viscosity solutions, vanishing viscosity, and why we need them.



The Cauchy-Type HJ Equation

Our chief object is the evolution (Cauchy) HJ equation

$$v_t(x, t) + H(t; x, \nabla_x v) = 0 \quad \text{in } \Omega \times (0, T], \quad v(x, 0) = v_0(x) \quad \text{in } \Omega,$$

with $H : (0, T] \times \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ continuous and g, v_0 **bounded, uniformly continuous (BUC)**.

- A closely related object is the **scalar conservation law / convection equation** $v_t + \sum_i f_i(v)_{x_i} = 0$, whose shock theory motivates the numerics.
- The HJ equation is **first-order, nonlinear, hyperbolic**.

First-order nonlinear hyperbolic PDEs generically fail to have smooth global solutions — the reason the entire viscosity-solution apparatus exists.



Why Classical Solutions Fail: Crossing Characteristics

- The **method of characteristics** integrates the PDE along curves in (x, t) .
- For nonlinear H , characteristics **cross**: multiple characteristics carry conflicting values to the same point $\rightarrow$ a **shock** / gradient discontinuity.
- Consequence: no global C^1 solution exists in general, **even when H and g are smooth**.
- Global analysis via classical PDE theory is "virtually impossible" (Crandall-Lions).

The value function of a differential game develops **kinks and shocks** right at the barrier — the most safety-relevant region. We must define a *weak* solution that survives there.



Viscosity Solutions¹

- **Viscosity solutions** are the correct weak solution class for HJ PDEs: they exist almost everywhere, and enjoy **existence, uniqueness, and stability** theorems.
- Definition (sketch) via test functions: v is a viscosity *subsolution* if for every smooth φ touching v from above at (x_0, t_0) , $\varphi_t + H(\cdot, \nabla \varphi) \leq 0$; *supersolution* with the reverse inequality and touching from below; a **solution** is both.
- This selects the physically-correct kinked solution and discards spurious ones.

Viscosity theory gives us a **unique** object to compute and to certify — indispensable for a safety guarantee.

¹ Crandall & Lions, 1983



Vanishing Viscosity

Introduce $\delta > 0$ and regularize (parabolic smoothing, as in gas dynamics):

$$v_t^\delta + H(t; x, \nabla v^\delta) = \frac{\delta}{2} \Delta v^\delta \quad \text{in } \Omega \times (0, T], \quad v^\delta(x, 0) = g(x).$$

- The added Laplacian makes the PDE **parabolic**, hence classically well-posed: $v^\delta \in C^{2,1}$, unique, stable, BUC for all T .
- Traversing the limit $\delta \rightarrow 0$ recovers the unique viscosity solution.

δ is the linchpin of HJ-Gauss: it is both the **smoothing** that admits a classical solution *and* the **diffusion coefficient** of the heat equation we are about to expose.



The Crandall-Lions Error Bound

The viscous solution approximates the inviscid viscosity solution at a known rate:

$$\sup_{t \in (0, T]} \sup_{x \in \mathbb{R}^n} |v(t, x) - v^\delta(t, x)| \leq k\sqrt{\delta}, \quad k > 0.$$

- **Bias scales as $O(\sqrt{\delta})$:** smaller $\delta \rightarrow$ more accurate.
- But (preview) smaller $\delta \rightarrow$ more concentrated exponential weights $\rightarrow$ higher Monte-Carlo variance.
- This is the **bias-variance knob** we will formalize in Part 7 ($\delta \sim N^{-1/3}$).

One scalar δ trades geometric accuracy against sampling variance. Choosing it well is the practical art of the method.



Admissible Data on $\mathbb{R}^n$

- Whole-space heat theory requires the datum in $C(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$.
- The transformed datum is $\omega^\delta(0, \cdot) = e^{-cg} - \text{not } g$ itself. This matters: the signed distance ℓ is *unbounded above*, but for a bounded target $g \geq g_{\min}$ and $g \rightarrow +\infty$ as $|x| \rightarrow \infty$, so

$$0 < e^{-cg(y)} \leq e^{-cg_{\min}}, \quad e^{-cg(y)} \rightarrow 0 \text{ as } |y| \rightarrow \infty.$$

- Hence e^{-cg} is continuous, strictly positive, bounded, decaying $\rightarrow$ the heat solution is the **unique bounded** one, the integral converges absolutely, and the Gaussian expectation is genuine and **unbiased**.

Only the **lower** bound $g \geq g_{\min}$ is needed; no upper bound on g is used. This is what lets us sample over all of $\mathbb{R}^n$.

Part 2

Reachability & Safety Foundations

What a certified safety set *is*, before we compute one.

What Is Reachability?

- **Reachability** concerns the *decidability* of a dynamical system's trajectory evolution across a phase space.
- A reachable system is **decidable** when one can compute *all* states reachable from an initial condition **in a finite number of steps**.
- Dual questions:
 - **Forward**: where can the system go from here?
 - **Backward**: from which states is a target inevitably reached (or avoidable)?
- Safety analysis is naturally **backward**: characterize the set of states doomed to enter a danger set, then stay out of it.

Certifying a learned controller, neural policy, or planner means proving it satisfies all specified requirements — a **verification** problem. Reachability is the geometric engine of that verification.

Backward Reachable Sets and Tubes

- **Backward Reachable Set (BRS)**: states that reach the target at a *specific* time.
- **Backward Reachable Tube (BRT)**: states that reach the target at *some* time in $[0, T]$ — the safety-relevant object.
- **Reach-Avoid Tube (BRAT)**: states that can reach a goal while avoiding an obstacle set.
- **Robustly-Controlled BRT (RCBRT)**: when the controller must counter a **worst-case disturbance** — the adversarial guarantee.

$$\mathcal{L}([-T, 0], \mathcal{L}_0) = \{x \in \mathbb{R}^n : \exists \beta \in \mathcal{B}(t) \forall u \in \mathcal{U}(t), \exists \bar{\tau} \in [-T, 0], \xi(\bar{\tau}) \in \mathcal{L}_0\}.$$

The strategy structure $(\exists \beta \forall u)$ encodes the game: disturbance plays a nonanticipative strategy β against control u .

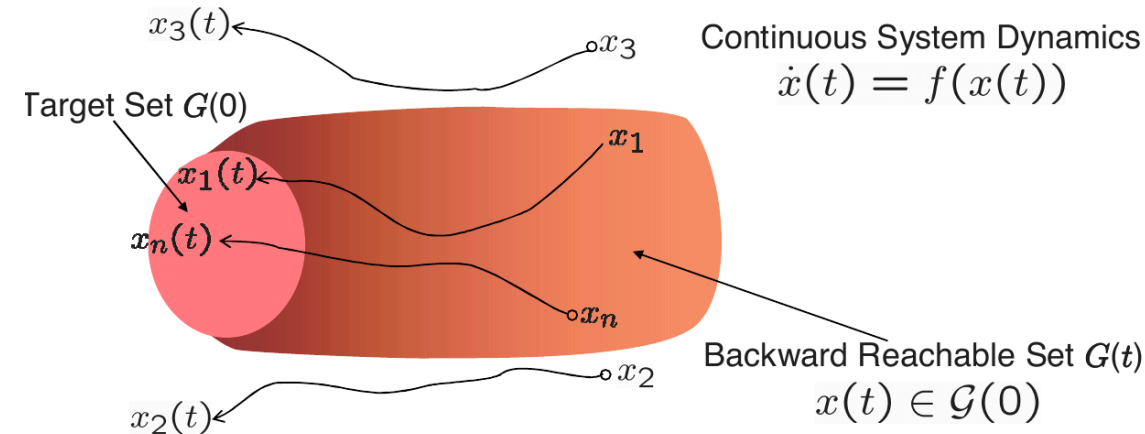


Why "Backward" Reachable Sets

- A **continuous backward reachable set** is the set of all states from which trajectories can reach a given target set $G(0)$.
- Called "**backward**" to distinguish it from the *forward* reachable set.
- **To compute it, run the dynamics backward in time** from the target set.

For safety, the target is usually the *unsafe* set, so the backward reachable set is the set of states doomed to become unsafe.

(Mitchell 2004, Reach Sets and the HJ Equation)

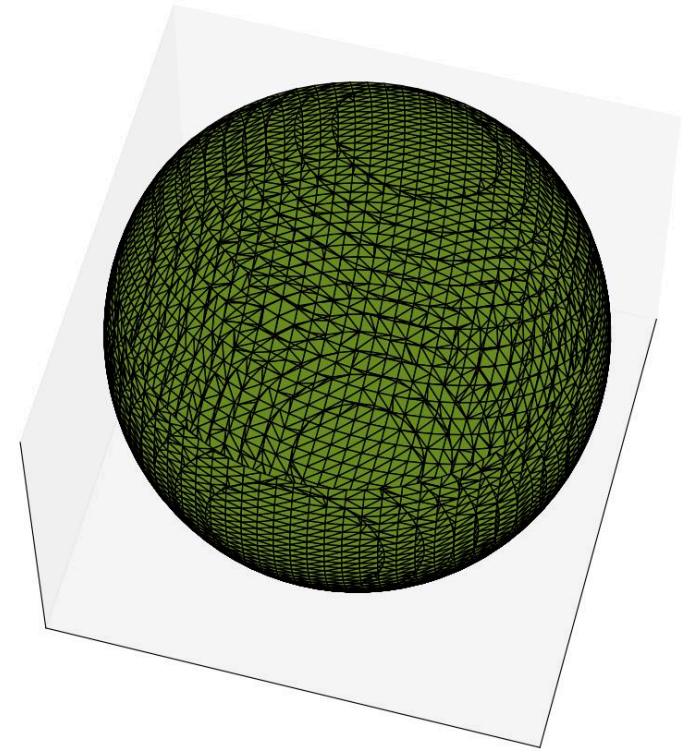


How to *Represent* the Set/Tube?

Computing a reachable set poses two coupled problems:

- **Represent** the set of reachable states.
- **Evolve** that set according to the dynamics.

Inset: A sphere as an SDF — LevelSetPy (L. Molu), ACM TOMS 51(2), 2025.



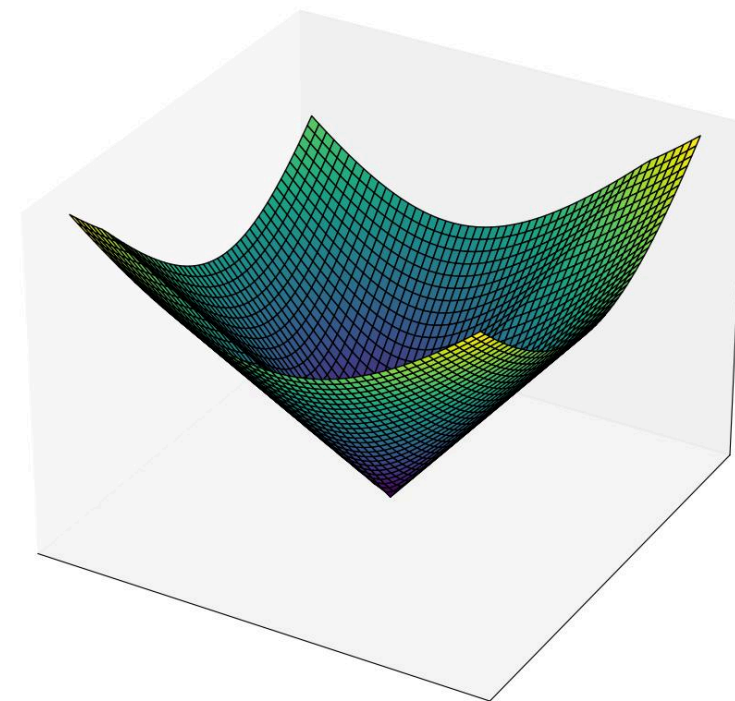
Implicit Surface Functions

Level-set idea: Represent a set $G(t)$ as an **isosurface of a scalar function** $\phi(x, t)$ ¹:

- **State-space dimension does not matter conceptually**: the same ϕ machinery works in any n .
- Surfaces **automatically merge and separate** as the set evolves.
- **Geometric quantities** (normals, curvature, distance) are easy to compute from ϕ .

Critical talk juncture: the reachable set is the zero sublevel set of a value function ϕ .

Inset: Set union of two spheres — LevelSetPy (L. Molu), ACM TOMS 51(2), 2025. · ¹ Osher & Sethian



How to *Represent* the Set/Tube?

For continuous systems $\dot{x} = f(x)$, two families:

- **Lagrangian** (forward sets; restricted dynamics/shapes; overapproximation): HyTech, Checkmate, d/dt , ellipsoidal¹.
- **Eulerian** (backward sets; general dynamics incl. competitive inputs; **implicit** set representation).
 - **HJ-Gauss** belongs here.

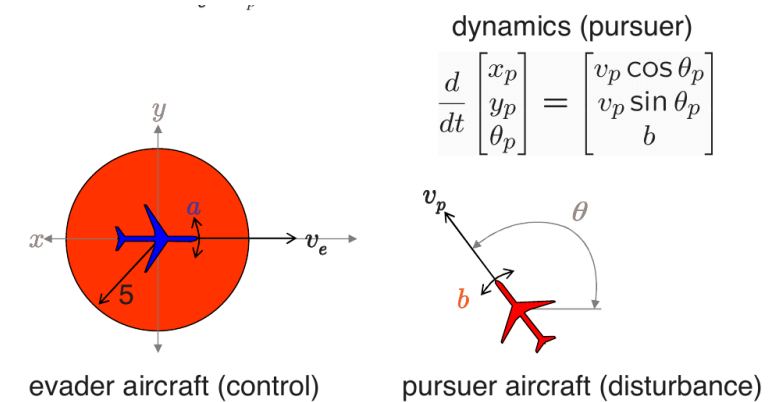
¹ Kurzhanski

Canonical Example: Two Identical Vehicles

Classical collision avoidance:

- Collision if the vehicles come within **5 units** of each other.
- **Evader** picks turn rate $|a| \leq 1$ to *avoid*;
- **Pursuer** picks $|b| \leq 1$ to *cause* collision; fixed equal speeds $v_e = v_p = 5$.

(Mitchell 2004, Reach Sets and the HJ Equation)



24 Oct 04

Ian Mitchell (UBC Computer Science)

11



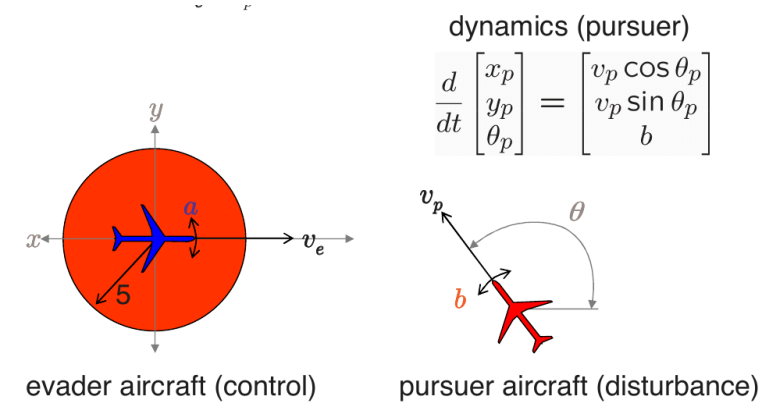
Canonical Example: Two Identical Vehicles

Classical collision avoidance:

- Work in **relative coordinates** with the evader fixed at the origin
- State is relative position (x, y) and relative heading ψ .

NB: Same relative-coordinate pursuit-evasion (PE) game we solve with HJ-Gauss (Part 8); and the pairwise MAPF conflict of Part 9.

(Mitchell 2004, Reach Sets and the HJ Equation)



24 Oct 04

Ian Mitchell (UBC Computer Science)

11



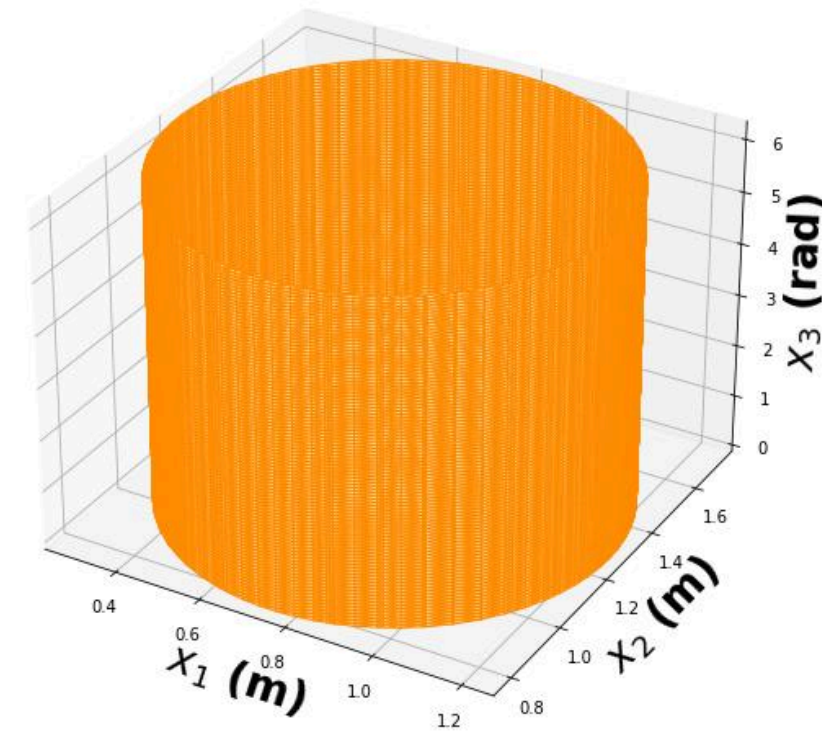
Evolve: The Time-Dependent HJ Equation

The set evolves by a (modified) Hamilton-Jacobi PDE:

- A **first-order hyperbolic PDE** whose solution can form **kinks** (discontinuous derivatives).
- The right weak solution is the **viscosity solution**¹.

(Mitchell 2004, Reach Sets and the HJ Equation); viscosity:
Crandall-Evans-Lions; level sets: Osher-Sethian · Dubins BRT:
LevelSetPy (L. Molu), ACM TOMS 51(2), 2025 · ¹ Crandall, Evans,
Lions

BRT at 0.00 secs.



Evolve: The Time-Dependent HJ Equation

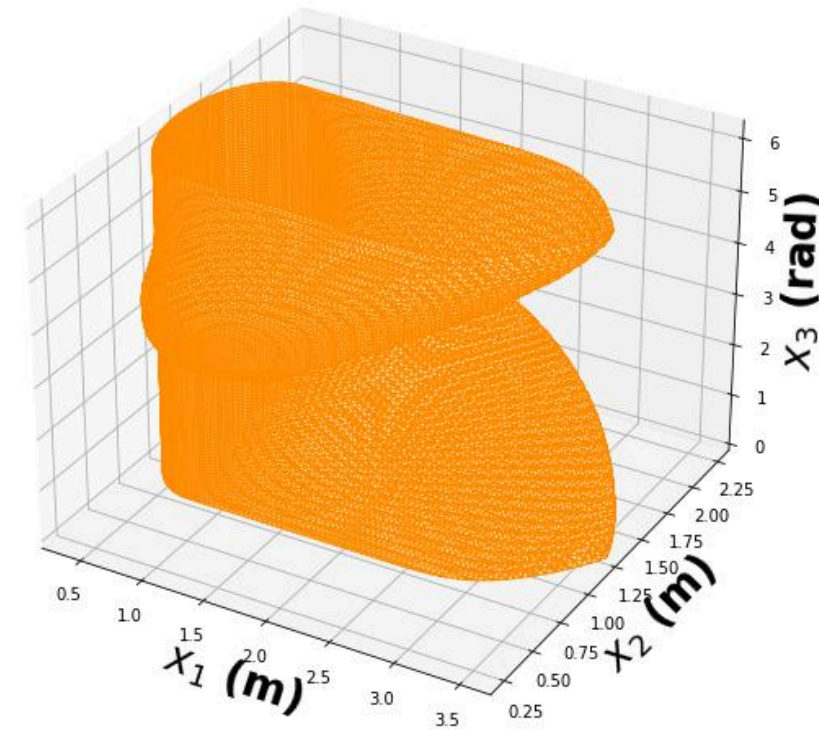
The set evolves by a (modified) Hamilton-Jacobi PDE:

- **Level-set methods** produce convergent numerical schemes¹:
 - Non-oscillatory high-accuracy spatial derivatives;
 - A stable/consistent numerical Hamiltonian; and
 - Total variation-diminishing (TVD) high-order explicit time integration.

This is the classical grid pipeline of Part 3. HJ-Gauss tackles this computational cost.

(Mitchell 2004, Reach Sets and the HJ Equation); viscosity:
Crandall-Evans-Lions; level sets: Osher-Sethian · Dubins BRT:
LevelSetPy (L. Molu), ACM TOMS 51(2), 2025 · ¹ Osher, Sethian

BRT at 2.50 secs.



The Game Value and the Optimal-Stopping Fix

- The **terminal-cost differential game**
 - Trajectories $\xi(\cdot; x, t, a, b)$;
 - **Value function** $\phi(x, t)$ is the viscosity solution of the HJ equation¹.

This modified/augmented Hamiltonian is the $\min\{0, \cdot\}$ freeze term on our HJI-RCBRT slide.

¹ Evans & Souganidis, 1984

The Game Value and the Optimal-Stopping Fix

- To stop trajectories from passing *through* the target $G(0)$ (so the set is a **tube**, not just a set),
 - **Augment the disturbance input;**
 - The augmented HJ equation solves for the reachable set;
 - The augmented Hamiltonian is the modified $\min\{0, H\}$ Hamiltonian.

This modified/augmented Hamiltonian is the $\min\{0, \cdot\}$ freeze term on our HJI-RCBRT slide.

Three Eulerian Approaches (All Equivalent)

The method sits among Eulerian formulations:

- **Static HJ**: Minimum time-to-reach¹; (dis)continuous implicit representation; yields optimal-input information.
- **Viability kernels**: Set-valued analysis for very general dynamics²; discrete implicit representation; overapproximation guarantee.
- **Time-dependent HJ** (this method): Continuous solution, optimal-input information throughout the state space, high-order accurate.

All three are **theoretically equivalent**; HJ-Gauss is a new solver for the time-dependent HJ formulation.

(Mitchell 2004, Reach Sets and the HJ Equation); Falcone/Sethian; Aubin/Saint-Pierre · ¹ Falcone; Sethian · ² Aubin; Saint-Pierre

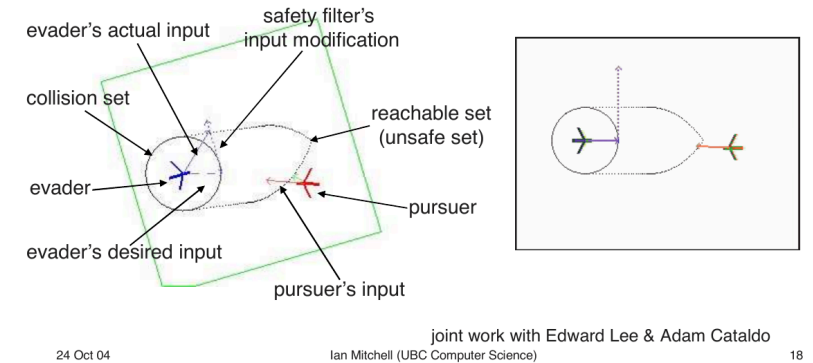
Why It Matters: Two Applications

Reachable sets already drive real-world safety systems:

- **Softwalls for aircraft safety:** Filter evader's input so the pursuer never enters the reachable (unsafe) set — a certified safety filter (with E. Lee & A. Cataldo).
- **Collision alert for ATC:** Flag aircraft pairs whose flight plans intersect and whose relative state enters the collision region.
 - A one-hour Oakland-airspace sample: **1590 pairs, 25 detected conflicts, 2 false alerts.**

These are the ancestors of the certified conflict predicate / shield we bring to warehouse MAPF in Part 9.

(Mitchell 2004, Reach Sets and the HJ Equation); Softwalls with E. Lee & A. Cataldo





Validation: Merz's Analytic Solution

- For **identical PE** dynamics, the reachable set admits an **analytic solution**¹.
- Merz placed the pursuer at the origin; the game is **not symmetric**.
- That analytic solution is used to **validate the numerical algorithm**² — the same discipline behind our LevelSetPy-vs-Monte-Carlo comparison in Part 8.

Takeaway of the primer: reachability = represent implicitly, evolve in a HJ PDE game, solve for the viscosity solution.

HJ-Gauss keeps this exactly and changes only *how* the PDE is solved.

(Mitchell 2004, Reach Sets and the HJ Equation); Merz (1972); Mitchell (2001) · ¹ Merz, 1972 · ² Mitchell, 2001



Target Set and Zero Level Set

- The **target set** at horizon T is the invariant set

$$\mathcal{L}_0(T) = \{x \in \mathbb{R}^n : v(0, x) \leq 0\},$$

robustly controlled over the **distance-to-target** cost $g(0, x)$.

- Numerically g is a **signed-distance function** $\ell(x)$ whose zero sublevel set is the target: negative inside, positive outside.
- The BRT is recovered as the **zero sublevel set** of the value function:

$$x(t) \in \mathcal{L}(\cdot) \iff v(t, x) \leq 0.$$

The Hamilton-Jacobi-Isaacs Value Function

The value function of the RCBRT is the **viscosity solution** of the variational HJ-Isaacs equation

$$v_t(t, x) + \min\{0, H(t; x, Dv)\} = 0, \quad v(0, x) = g(0, x),$$

with the **game Hamiltonian**

$$H(t; x, p) = \max_u \min_w \langle p, f(t; x, u, w) \rangle.$$

- The $\min\{0, \cdot\}$ **freeze** term makes the set only *grow* (a tube, not a set) — states already captured stay captured.
- For reach-avoid, the variational inequality couples the growth term with the obstacle datum ℓ .

This Hamiltonian is **positively 1-homogeneous in p , state-dependent, nonconvex, and sign-changing** across the barrier. Those four properties will rule out every convex-duality shortcut.

Reach vs Reach-Avoid; Sign & Time Conventions

- **Backward-reachability convention:** t is the **backward horizon** (time-to-go); the datum is posed at $t = 0$ and the tube grows over $(0, T]$.
- **Reach (capture) tube:** $v_t + \min\{0, H\} = 0$.
- **Reach-avoid tube (viscous):**

$$\min \left\{ v_t^\delta + H^\delta - \frac{\delta}{2} \Delta v^\delta, \quad g(t, x) - \ell(t, x) \right\} \leq 0.$$

- Physical time is $t_{\text{phys}} = T - t$; the datum is the terminal cost in physical time.

These conventions matter for signs during implementation and for *which* set (reach vs avoid) a negative value certifies. We adopt the backward-reachability viscosity-solution convention throughout.

Why This Is the Right Safety Object

- **Worst-case, not average-case:** the RCBRT certifies safety against *all* admissible disturbances — the guarantee a safety case needs.
- **Geometric and set-valued, not trajectory-valued:** it characterizes every unsafe initial condition at once, not one rollout.
- **Composable:** the zero level set can be intersected, unioned, and propagated; it plugs into supervisory control and shielding.
- **Certificate-grade:** a signed value with an error bound yields a decision — SAFE / UNSAFE / UNDETERMINED (Part 7).

The cost of this rigor is computational: solving the HJ(I) PDE. The rest of the talk is about paying that cost at scale.

Part 3

The Grid Pipeline: LevelSetPy

How Reachable Sets Are Computed Today — and Why It Is $O(M^n)$.

Molu. ACM TOMS 2025 · IEEE CDC 2024

TOMS: ACM Transactions on Mathematical Software · CDC: Conference on Decision and Control.

The Level-Set Idea¹

- Represent the reachable set implicitly as the **zero sublevel set** of $v(t; x)$ stored on a Cartesian grid.
- Evolve the interface by integrating the HJ PDE on the grid:
 - Discretize space (upwinding),
 - Stabilize the Hamiltonian (Lax-Friedrichs),
 - March in time (TVD-RK).

¹ Osher-Sethian

The Level-Set Idea¹

- **LevelSetPy** (our prior work) reimplements the 2004 MATLAB Level Set Toolbox in NumPy/CuPy, GPU-accelerated and interoperable with modern Python (PyTorch, SciPy, ROS).

MATLAB ToolboxLS (2004)	LevelSetPy (2024/25)
CPU-only, single-threaded	NumPy + CuPy GPU
Slow for modern problems	Fast, batched, portable
No modern-library plug-in	ROS/PyTorch/SciPy compatible

LevelSetPy makes grid reachability *fast*. It does not change the *memory scaling* — the theme of Part 4.

¹ Osher-Sethian

Implicit Surfaces & Signed-Distance Initialization

- The target set is initialized as a **signed-distance function** $\ell(x)$: negative inside, positive outside, $|\nabla \ell| = 1$.
- Stored as an n -dimensional array over a Cartesian grid: M points per axis $\rightarrow M^n$ cells.
- Geometric primitives (spheres, cylinders, half-spaces) and boolean set operations (union/intersection/complement) are implemented as min/max on the level-set field.
- Example: a capture ball of radius r is $\ell(x) = \|x\| - r$; a cylinder ignores the periodic θ axis.

The implicit representation is elegant and closed under set algebra — but it stores the value at **every** grid node.



Spatial Derivatives via Upwinding

- The co-state $p = \partial v / \partial x$ must be approximated from grid values with the correct **direction of information flow** (upwinding) to remain stable at shocks.
- **First-order upwinding:** one-sided differences chosen by the sign of the characteristic speed.
- Left/right approximations $D^- v$, $D^+ v$ feed the numerical Hamiltonian; the choice prevents differencing across a discontinuity.

$$v_x(x, t) \approx \frac{\partial v(x, t)}{\partial x} \quad (\text{one-sided, direction by wind}).$$

Naive centered differences ring and go unstable at the barrier; upwinding is the minimal fix, refined next by ENO/WENO.

ENO: Essentially Non-Oscillatory Reconstruction

- **ENO** picks, among candidate stencils, the **smoothest** one to interpolate the derivative — avoiding stencils that straddle a shock.
- Orders implemented: **ENO2** (2nd), **ENO3** (3rd), via `upwind_first_eno2.py`, `upwind_first_eno3.py`.
- Higher order → sharper interface, lower numerical diffusion, at more stencil work per node.

ENO chooses *one* smoothest stencil. WENO improves on this by blending stencils with adaptive weights.

WENO5: Weighted ENO Shock Capture

- **WENO5**¹ forms a **convex combination** of three candidate substencils rather than choosing one, achieving 5th-order accuracy in smooth regions and non-oscillatory capture at shocks.
- Substencils on the grid index i :

$$\{i - 3, \dots, i\}, \quad \{i - 2, \dots, i + 1\}, \quad \{i - 1, \dots, i + 3\},$$

combined with nonlinear weights that de-emphasize stencils crossing a discontinuity.

- Implemented in `upwind_first_weno5.py`.

WENO5 is the workhorse for accurate reachable-set boundaries — and its per-node cost is one reason grid solves are expensive even before the memory wall.

¹ Liu, Osher, Chan 1994

The Lax-Friedrichs Numerical Hamiltonian

- The analytic $H(x, p)$ is replaced by a **monotone numerical Hamiltonian** $\hat{H}(x, p^-, p^+)$ using the Lax-Friedrichs flux:

$$\hat{H} = H\left(x, \frac{p^- + p^+}{2}\right) - \frac{1}{2} \alpha \cdot (p^+ - p^-),$$

with dissipation coefficient $\alpha = \max |\partial H / \partial p|$ over the relevant range (dimension-wise for global LF).

- Monotonicity guarantees convergence to the viscosity solution.

The artificial dissipation α is a *numerical* cousin of the viscosity δ — both stabilize the barrier. HJ-Gauss will trade the grid's LF dissipation for the analytic δ .



Time Integration: TVD Runge-Kutta

Method of lines + **Total-Variation-Diminishing Runge-Kutta**¹, so the interface does not spuriously oscillate:

- **Forward Euler (1st):** $v^{n+1} = v^n + \Delta t L(v^n), L = -\hat{H}$.
- **TVD-RK2:** Euler step to $v^{(1)}$, second Euler step, then convex average $v^{n+1} = \frac{1}{2}v^n + \frac{1}{2}v^{(2)}$.
- **TVD-RK3:** three stages with convex-combination weights $(1), (3/4, 1/4), (1/3, 2/3)$.

TVD-RK preserves monotonicity of the spatial scheme in time. Each stage is a full grid sweep.

¹ Shu-Osher



The CFL Condition

- Explicit integration is stable only under a **Courant-Friedrichs-Lewy** step restriction:

$$\Delta t \leq \frac{\text{CFL}}{\sum_i \max |\partial H / \partial p_i| / \Delta x_i}, \quad \text{CFL} \in (0, 1).$$

- Numerical information must not travel more than one grid cell per step.
- Finer grids ($\Delta x \downarrow$) force **smaller** $\Delta t \rightarrow$ more steps $\rightarrow$ compounding the cost.

CFL couples spatial and temporal resolution: refining space to sharpen the barrier makes *both* the memory and the step count worse.

Package Anatomy & a Worked Example

Module	Role
<code>grids</code> , <code>initialconditions</code>	Cartesian grids, signed-distance shapes
<code>spatialderivative</code>	Upwind first, ENO2/3, WENO5
<code>explicitintegration</code>	Lax-Friedrichs Hamiltonians, TVD-RK, CFL
<code>dynamicalsystems</code>	Vehicle models (Dubins, rockets)
<code>visualization</code>	Isosurface / marching cubes rendering

- **Worked example:** the rockets-launch pursuit-evasion game; 2D (x, z) slices of the 3D relative-state BRT, evolved backward over $(0, T]$.
- Multi-agent verification: flocks/murmurations partitioned into per-flock games.

A clean, tested, GPU-portable stack — the state of practice we now try to move beyond.

Grid Complexity: The Accounting

- **Memory:** $O(M^n)$ — store the value (and gradient) at every node.
- **Time per step:** $O(M^n)$ node updates $\times$ stencil width $\times$ RK stages, under a CFL-bounded step count.
- GPU acceleration cuts the *constant* and parallelizes node updates, but the **exponent n is untouched**.

$$n = 6, M = 100 \Rightarrow 10^{12} \text{ cells} \approx 8 \text{ TB per double array.}$$

This is the wall. Part 4 quantifies it and surveys who has tried to climb it.

Part 4

The Wall

Curse of dimensionality, and who has tried to climb it.

The Curse of Dimensionality, Quantified

Grid memory is $O(M^n)$ with M points per dimension:

State dim n	$M = 100$ cells	Feasibility (double array)
2	10^4	✓ trivial
3	10^6	✓ easy
4	10^8	⚠ ~0.8 GB, heavy
5	10^{10}	✗ ~80 GB
6	10^{12}	✗ ~8 TB
45 (our multi-agent game)	10^{90}	✗ exceeds atoms in the observable universe

- Even a coarse $M = 101$ grid at $n = 45$ needs $101^{45} \approx 10^{90}$ cells.
- GPU acceleration lowers wall-clock, **not** the exponent.

Reachability's most valuable regime — many interacting agents, rich dynamics — is where grids are impossible.



Attempt 1: PINN Reachability (DeepReach)

- **DeepReach** trains a neural network to minimize the HJ PDE residual directly (physics-informed loss), no reference grid.
- **Pro:** Trades grid storage for network weights; sidesteps $O(M^n)$ memory.
- **Con:** Scales only to moderate dimensiond (reported ~9-10D);
 - And **accuracy degrades** as n grows; training is a nonconvex optimization with no viscosity-solution guarantee.

Catch-22: Learning-based solvers help, but do not deliver a certified, dimension-robust representation.
HJ-Gauss keeps the viscosity-solution semantics.



Attempt 2: Convex-Duality Formulas (Hopf / Lax-Oleinik)

HJ-Gauss trades their exactness for generality, at the quantified price of a linearization residual bound (Part 7).

- **Hopf** and **Lax-Oleinik** formulas¹ evaluate the value **pointwise** via convex optimization — grid-free at a point.
- **Limitation:** they require a **convex** (or state-independent) Hamiltonian.
- Reachability's $H(x, p) = \max_u \min_w \langle p, f \rangle$ is **state-dependent and nonconvex** → outside this class.

It handles the nonconvex, state-dependent Hamiltonian head-on.

¹ Darbon-Osher 2016; Chow-Darbon-Osher-Yin 2017; Kirchner 2018

Attempts 3-4: Decomposition, Path Integral, Sampling

Family	Idea	Boundary
System decomposition ¹	Remove dimension by splitting self-contained subsystems	Only when coupling structure permits; HJ-Gauss is indifferent to coupling and <i>complements</i> it
Stochastic PDE / path integral ²	Log-substitution linearizes the stochastic HJB under noise-control duality	Diffusion comes from <i>process noise</i> ; ours is a deterministic worst-case game , δ an analysis parameter
Sample-based stochastic reachability ³	Certify probabilistic reach-avoid	We certify adversarial worst-case reachability — a different, stronger guarantee

The open gap: a grid-free, storage-light scheme for **nonconvex, state-dependent, adversarial** Hamiltonians. That is HJ-Gauss.

¹ Chen-Herbert · ² Kappen; Theodorou · ³ Summers-Lygeros; Lesser-Oishi



Where HJ-Gauss Sits

- Keeps the **viscosity-solution** semantics (unlike PINN heuristics).
- Handles **nonconvex, state-dependent** H (unlike Hopf/Lax-Oleinik).
- **Indifferent to coupling structure** (complements decomposition).
- Certifies **adversarial worst-case** safety (unlike probabilistic sampling).
- Memory $O(N \cdot n)$, grid- and discretization-free.

The rest is *how*: a Cole-Hopf transformation, a Feynman-Kac expectation, and a frozen-coefficient Picard iteration.

Part 5

HJ-Gauss Core Theory

Cole-Hopf Transformation → Linear Heat Equation



→ Feynman-Kac Formula → Picard Iteration.

The Theoretical Roadmap of the Method

1. Start from the **viscous** HJ PDE (Part 1).
2. Apply a generalized **Cole-Hopf** transform $\omega = e^{-cv} \rightarrow$ a **Linear heat equation**.
3. Solve the heat equation by its **Gaussian heat kernel** $\rightarrow$ a **Feynman-Kac expectation**.
4. Recover value (log-sum-exp) and gradient from that expectation.
5. Because c depends on the unknown Dv , **iterate**: freeze, solve, update (Picard) = **Algorithm 1**.
6. Analyze exactness (quadratic case) vs the **quasi-linearization residual** (general case).

Each arrow is a theorem or proposition in the paper; we walk them one by one.

The Transformation, and the Closed-Form/Approximate Split

For a general H , define the spatially-varying coefficient

$$c(t; x) = \frac{2}{\delta} \cdot \frac{H^\delta}{|Dv^\delta|^2}, \quad H^\delta := H(t; x, Dv^\delta),$$

and set $\omega^\delta := \exp(-c v^\delta)$.

- **Quadratic case** $H = \frac{1}{2}|p|^2$: $c = 1/\delta$ constant, ω^δ solves the **homogeneous heat equation** with **zero residual** — a genuine Cole-Hopf identity.
- **General H** : the transform induces a residual $R = R_{\text{alg}} + R_{\text{der}}$;
 - c 's choice eliminates the **algebraic** part R_{alg} , leaving the **derivative** part R_{der} .

No freezing of c 's *values* removes its *derivatives*. Hence for nonlinear H this is a **quasi-linearization**, not an identity — and the leftover R_{der} is the residual we will bound.

Picard Quasi-Linearization: The Surrogate View

- At iteration k , **freeze** $c^{(k)}$ at the current iterate and solve the linear heat equation **in closed form**.
- Equivalently: replace the true Hamiltonian by the **locally-matched quadratic surrogate**

$$\tilde{H}^{(k)}(p) = \frac{\delta}{2} c^{(k)} |p|^2,$$

solve the resulting viscous HJ equation in closed form, obtain $v^{(k+1)}$, update $Dv^{(k+1)}$ and $c^{(k+1)}$.

- **The iteration's limit is the fixed point of the surrogate solve map** — equal to the viscous solution *only* when $R_{\text{der}} \equiv 0$ (the quadratic case).

"Successive quadratic matching": each step is a locally closed-form quadratic-Hamiltonian solve, refined by re-estimating the co-state.

Reduction to the Heat Equation (Proposition)

With $c^{(k)}$ frozen, ω^δ solves the heat initial-value problem

$$\omega_t^\delta - \frac{\delta}{2} \Delta \omega^\delta = 0 \quad \text{in } \mathbb{R}^n \times (0, T], \quad \omega^\delta(0, x) = \exp(-c g(x)).$$

- When $H = \frac{1}{2}|p|^2$, $c = 1/\delta$: **closed-form, no residual**.
- For general H : discards the derivative residual (Lemma, appendix) — the quasi-linearization.

A nonlinear, nonconvex, state-dependent HJ PDE has become **the heat equation** — the most classical linear parabolic PDE, with an explicit Gaussian solution.

The Gaussian Heat Kernel & Feynman-Kac

The heat IVP has the unique bounded Green's-convolution solution

$$\omega^\delta(t, x) = \frac{1}{(\sqrt{2\pi\delta t})^n} \int_{\mathbb{R}^n} e^{-\frac{|x-y|^2}{2\delta t}} e^{-c g(y)} dy,$$

which, by **Feynman-Kac**, is a **Gaussian expectation**:

$$\omega^\delta(t, x) = \mathbb{E}_{y \sim \mathcal{N}(x, \delta t I_n)} [e^{-c g(y)}].$$

- The free-space kernel integrates to one over $\mathbb{R}^n$ (and no proper subset) — every estimator is posed on all of $\mathbb{R}^n$.
- Sampling $y \sim \mathcal{N}(x, \delta t I_n)$ and averaging $e^{-c g(y)}$ is an **unbiased** estimator of ω^δ .

The solution is a **Gaussian roll-out average of the terminal cost**. This is the entire computational payload — no grid, no marching.

Recovering the Value (Lemma)

Invert the transform:

$$v^\delta(t, x) = -\frac{1}{c^{(k)}} \log \mathbb{E}_{y \sim \mathcal{N}(x, \delta t I_n)} [e^{-c^{(k)} g(y)}].$$

- A **log-sum-exp** of Gaussian-sampled terminal costs.
- Numerically evaluated in a **log-domain-stable** form (log-mean-exp) to avoid overflow/underflow.
- M arbitrary evaluation states $x_1, \dots, x_M$ — laid on a grid or scattered — each with its own fresh N samples.

The value at a query point is a scalar reduction over N samples. Embarrassingly parallel across both M states and N samples.

Recovering the Gradient (Corollary)

The spatial gradient (co-state) has a closed Gaussian-expectation form:

$$Dv^\delta = \frac{1}{t \delta c^{(k)}} \left(x - \frac{\mathbb{E}_{y \sim \mathcal{N}(x, \delta t I_n)}[y e^{-c^{(k)} g(y)}]}{\mathbb{E}_{y \sim \mathcal{N}(x, \delta t I_n)}[e^{-c^{(k)} g(y)}]} \right).$$

- A **weighted-mean shift**: the gradient points from x toward the exponentially-weighted centroid of the samples.
- No finite differencing, no stored field — the co-state is a byproduct of the same samples.

This gradient is (up to constants) a **score function** $\nabla \log \omega$. Hold that: it is the entire bridge to diffusion in Part 10.

Admissible Data & Unbiasedness (Recap as a Guarantee)

- Datum $\omega^\delta(0, \cdot) = e^{-cg}$ is continuous, strictly positive, bounded above by $e^{-cg_{\min}}$, decaying at infinity.
- $\Rightarrow$ the heat solution is the **unique bounded** one; the integral converges absolutely; the Gaussian expectation is **genuine and unbiased**.
- Only $g \geq g_{\min}$ is required (bounded target); no upper bound on g .

The estimator drawing $y \sim \mathcal{N}(x, \delta t I_n)$ over all of $\mathbb{R}^n$ is **unbiased for the quantity inside the logarithm** — the foundation for the concentration bound of Part 7.

Sign and the Removable Zero of the Coefficient

- $c \sim C_H / (\delta |Dv^\delta|)$: it **grows without bound** in flat regions and **vanishes and flips sign** where optimal dynamics run tangent to the level set (the barrier).
- The zero of c is a **removable** singularity of the estimator: as $c \rightarrow 0$,

$$-\frac{1}{c} \log \mathbb{E}[e^{-cg}] \rightarrow \mathbb{E}[g],$$

i.e. the log-sum-exp gracefully degenerates to the **Gaussian mean** of the terminal data — the correct pure-diffusion limit ($H \rightarrow 0$ reduces the PDE to the heat equation).

- In implementation: evaluate in a **log1p**-stable form and **clip** $|c|$ to $[c_{\min}, c_{\max}]$.

The coefficient's pathologies are analytically understood and numerically tamed — clipping is itself a bounded, priced perturbation (next slide).

Coefficient Regularization (Lemma)

Run Algorithm 1 with the **regularized coefficient**

$$c_\eta(t; x) = \frac{2}{\delta} \cdot \frac{H^\delta}{|Dv^\delta|^2 + \eta},$$

equivalent to running it unregularized for the **perturbed Hamiltonian** $H_\eta = H |Dv^\delta|^2 / (|Dv^\delta|^2 + \eta)$, which satisfies a **uniform** bound

$$\|H_\eta - H\|_\infty \leq \frac{C_H \sqrt{\eta}}{2},$$

so the induced value-function perturbation is at most $TC_H \sqrt{\eta}/2$.

Regularization cures the flat-interior blow-up at a **quantified, uniform** price — no hidden instability.

Algorithm 1: Quasi-Linearization (Cole-Hopf)

$$\text{Fix } \varepsilon > 0, \quad v^{(0)}(t, x) = g(x), \quad c^{(0)} = \frac{2 H(t, x, Dg)}{\delta |Dg|^2}.$$

for $k = 0, 1, 2, \dots$:

(1) freeze $c^{(k)}$ at the current iterate;

(2) solve $\omega_t = \frac{\delta}{2} \Delta \omega$, $\omega^{(k)}(0, x) = e^{-c^{(k)} g(x)}$;

$$\Rightarrow \omega^{(k+1)}(t, x) = \mathbb{E}_{y \sim \mathcal{N}(x, \delta t I)} [e^{-c^{(k)} g(y)}] \quad (\text{Monte-Carlo});$$

(3) recover $v^{(k+1)} = -\frac{1}{c^{(k)}} \log \omega^{(k+1)}$;

(4) update $Dv^{(k+1)}$, $c^{(k+1)} = \frac{2 H(t, x, Dv^{(k+1)})}{\delta |Dv^{(k+1)}|^2}$ clip to $[c_{\min}, c_{\max}]$;

(5) stop when $\|v^{(k+1)} - v^{(k)}\| / \|v^{(k)}\| < \varepsilon$.

- Steps (2)-(4) are all **Gaussian-expectation reductions** over N samples.
- Convergence typically in ≤ 20 iterations (empirically 12-15).

Every line is grid-free. The only state carried between iterations is the coefficient field $c^{(k)}$ at the M evaluation states.

Evaluation States vs Monte-Carlo Samples (Remark)

- N = i.i.d. Gaussian draws $y_1, \dots, y_N \sim \mathcal{N}(x, \delta t I_n)$ used **independently at each** evaluation state x to form the estimators.
- Resampled **fresh** at every evaluation state and every Picard iteration.
- M = the evaluation states $x_1, \dots, x_M$ where v is queried — carry **no randomness**, may be a uniform grid or scattered arbitrarily.
- Total randomness per iteration: $M \times N$ **draws**.
- The zero level set is recovered by evaluating v^δ at the M states and isocontouring at level zero (marching cubes) — a deterministic post-process consuming **no further samples**.

The **solve is grid-free**; the grids that appear in results are *evaluation windows* for visualization and error measurement, not sampling constraints.

Memory: $O(N \cdot n)$ vs $O(M^n)$

	Grid level sets	HJ-Gauss
Representation	Stored field on M^n grid	N Gaussian samples per query, n -vectors
Memory	$O(M^n)$	$O(N \cdot n)$
Discretization	Cartesian grid + marching cubes to solve	None (grid-free solve)
Query states	Tied to grid	Arbitrary / scattered
Parallelism	Grid array ops	Over M states and N samples

- Example: $N = 14,000$, $n = 3 \rightarrow \approx 0.6$ MB/iteration; $n = 45 \rightarrow \approx 7.2$ MB/iteration (vs 10^{90} grid cells).

Memory now scales with **sample budget and dimension**, not with a grid raised to the dimension. This is the whole contribution in one line.

Part 6

Importance Sampling & Variance Control

Making the Gaussian estimator work in high dimensions.

The Weight-Degeneracy Problem

- The estimators are a **ratio of exponential-weight expectations**, e.g. the gradient's numerator/denominator both carry e^{-cg} .
- In high dimensions (or small δ), $c g$ is large over the kernel support, so the denominator $\mathbb{E}[e^{-cg}]$ is carried by a **rare event**.
- The weights collapse onto a few samples $\rightarrow$ the ratio's **variance explodes**; the effective sample size crashes.

Naive Gaussian sampling is unbiased but can be catastrophically high-variance right where we need it (high n , small δ). We fix it with importance sampling.

Tilted Proposal & Density-Ratio Reweighting

- Replace $y \sim \mathcal{N}(x, \sigma^2 I_n)$ ($\sigma^2 = \delta t$) by a **shifted** proposal $q_\theta = \mathcal{N}(x + \theta, \sigma^2 I_n)$.
- Reweight by the **density ratio**

$$w_\theta(y) = \exp\left(\frac{|\theta|^2 - 2\langle y - x, \theta \rangle}{2\sigma^2}\right),$$

which leaves **every expectation unbiased**.

We move the samples to where the integrand has mass, then correct via the ratio. Bias is untouched; variance can collapse by orders of magnitude.

The Zero-Variance Shift Is One Gradient Step

- The zero-variance proposal is proportional to the integrand $\varphi_x(y) e^{-cg(y)}$ itself.
- Its first-order **Laplace / Gaussian** approximation gives the shift

$$\theta^* = -\sigma^2 c^{(k)} Dv^{(k)}(t, x).$$

- That is, samples are pushed **one preconditioned gradient step** along the descent direction of the running value iterate.

The previous Picard iterate hands the sampler its drift for free. Reachability and importance sampling become the *same* computation — the value gradient is the optimal sampling direction.

Effective Sample Size as a Self-Diagnostic

- Monitor the **effective sample size** of the self-normalized weights $\tilde{w}_i = w_\theta(y_i)e^{-cg(y_i)}$:

$$\text{ESS} = \frac{(\sum_i \tilde{w}_i)^2}{\sum_i \tilde{w}_i^2} \in [1, N].$$

- $\text{ESS} \approx N$: weights well-spread $\rightarrow$ estimate certified.
- $\text{ESS} \rightarrow 1$: **weight degeneracy** $\rightarrow$ flag, increase N , raise δ , or re-tilt.

ESS is a free, per-query health check: it tells you *when* to trust the estimate and *why* it failed when it does.

Part 7

Guarantees

Concentration · Contraction · Residual · Total error · Certificate · Robustness.

▴ Four Error Sources, One Budget

The distance from the computed field $\hat{v}^{K,N,\delta}$ to the true inviscid viscosity solution v decomposes into four independently-controlled sources:

Source	Controlled by	Scaling
Iteration (Picard truncation at K)	More iterations	$C_1 \rho^K$
Monte-Carlo (finite N)	More samples	$C_2 N^{-1/2} \sqrt{\log(1/\delta_p)}$
Quasi-linearization residual $\mathcal{E}_{\text{ql}}$	Surrogate fidelity (not effort)	Floor
Viscosity (finite δ)	Smaller δ	$C_3 \sqrt{\delta}$

Three of four shrink with computational effort; the **residual is a property of the surrogate** and sets the floor. We now bound each.

▴ Finite-Sample Concentration (Theorem)

Fix phase (t, x) and frozen $c > 0$; assume $g_{\min} \leq g \leq g_{\max}$ on the sampling support. With $Z = e^{-cg(\zeta)}$, $\mu = \mathbb{E}[Z]$, $v_c = -\frac{1}{c} \log \mu$, $\alpha = e^{-cg_{\max}}$, $\beta = e^{-cg_{\min}}$, and $\hat{v}_{c,N} = -\frac{1}{c} \log \bar{Z}_N$:

$$\mathbb{P}(|\hat{v}_{c,N} - v_c| \geq \varepsilon) \leq 2 \exp\left(-\frac{2N\mu^2(1 - e^{-c\varepsilon})^2}{(\beta - \alpha)^2}\right).$$

- Bounded weights $\alpha \leq Z_i \leq \beta \Rightarrow$ Hoeffding.
- Rate is the standard $O(N^{-1/2})$, **independent of state dimension n** — the crux of the scalability claim.

Dimension enters the cost *per sample*, not the *number of samples* for a target accuracy. That is how we beat $O(M^n)$.

Explicit Sample Size, and Its Worst Case (Corollary + Remark)

- Since $\mu \geq \alpha$, it suffices to take

$$N \geq \frac{(\beta - \alpha)^2}{2\alpha^2(1 - e^{-c\varepsilon})^2} \log \frac{2}{\alpha}$$

to guarantee error $\geq \varepsilon$ with probability $\leq \alpha$.

- **Worst-case caveat:** with $c = 1/\delta$, $\beta/\alpha = e^{(g_{\max} - g_{\min})/\delta}$, so this bound grows like $e^{2(g_{\max} - g_{\min})/\delta}$ — **exponential in $1/\delta$** , while viscosity error is only $O(\sqrt{\delta})$.

The accuracy gain from shrinking δ and the sample cost pull **in opposite directions**. This Hoeffding bound is loose; the next slide tightens it, and experiments sit far below it.

Tightening: Bernstein & Jensen

- **Bernstein:** when $\text{Var}(Z) \ll (\beta - \alpha)^2/4$, a variance-aware tail bound gives strictly tighter concentration than Hoeffding.
- **Jensen:** the loose lower bound $\mu \geq \alpha = e^{-c g_{\max}}$ can be replaced by $\mu \geq e^{-c \mathbb{E}[g(\zeta)]}$, substantially reducing the required N .
- **Importance sampling** (Part 6) attacks the same variance directly.

The pessimistic $e^{1/\delta}$ is a worst-case artifact of Hoeffding + crude μ bound. In practice $N \in [14\text{k}, 20\text{k}]$ suffices at $\delta \in [0.08, 0.1]$.

The Iteration Operator (Contraction Setup)

- Fix evaluation states $x_1, \dots, x_M$ and time t ; let $\mathcal{G}$ be a stable gradient-reconstruction operator, $p_m(v) = (\mathcal{G}v)_m$.
- Coefficient-update map: $(\Gamma(v))_m = 2H(t, x_m, p_m(v)) / (\delta |p_m(v)|^2)$.
- Frozen-coefficient heat-kernel map:

$$(\Phi(c))_m = -\frac{1}{c_m} \log \mathbb{E}_{\zeta \sim \mathcal{N}(x_m, \delta t I_n)} [e^{-c_m g(\zeta)}].$$

- Algorithm 1 is the iteration $v^{(k+1)} = \Lambda(v^{(k)})$, $\Lambda = \Phi \circ \Gamma$.

The whole method is a fixed-point iteration of a composed map on $\mathbb{R}^M$ with the sup-norm. Contraction $\Rightarrow$ convergence.

Assumptions for Contraction (Assumption A)

On a closed admissible set $\mathcal{A}$ (sup-norm), assume:

1. $|g| \leq G$;
2. non-degenerate gradient & bounded coefficient: $m_0 \leq |p_m| \leq P_*$, $c_{\min} \leq (\Gamma v)_m \leq c_{\max}$;
3. Hamiltonian Lipschitz in co-state: $|H| \leq H_*$, $|H(\cdot, p) - H(\cdot, q)| \leq L_H |p - q|$;
4. reconstruction Lipschitz: $\|\mathcal{G}v - \mathcal{G}w\|_\infty \leq L_D \|v - w\|_\infty$;
5. invariance + contraction constant

$$q = \frac{2G}{c_{\min}} \cdot \frac{2L_D}{\delta} \left(\frac{L_H}{m_0^2} + \frac{2H_*P_*}{m_0^4} \right) < 1.$$

Non-degeneracy (item 2) holds on the **tube band** $\{|v^\delta| \leq \eta\}$ where the gradient is bounded away from zero — where the barrier lives. Outside, use the regularized coefficient.

Contraction Convergence (Theorem)

Λ is a **contraction** on $\mathcal{A} \Rightarrow$ unique fixed point $v^\star$, and for any $v^{(0)} \in \mathcal{A}$:

$$\|v^{(k+1)} - v^\star\|_\infty \leq q \|v^{(k)} - v^\star\|_\infty \leq q^{k+1} \|v^{(0)} - v^\star\|_\infty,$$

with the **a posteriori** estimate

$$\|v^{(k)} - v^\star\|_\infty \leq \frac{q}{1 - q} \|v^{(k)} - v^{(k-1)}\|_\infty.$$

- Iteration count for tolerance ε : $K = O(\log(1/\varepsilon))$ — **geometric** convergence.

The residual you *watch* (successive-iterate change) bounds the error you *cannot* see (distance to $v^\star$).
That is the a posteriori estimate — a runtime stopping certificate.

Scope of the Contraction (Remark)

- The theorem is about the **frozen-coefficient numerical map**, not the original nonlinear PDE.
- The fixed point v^* solves the viscous HJ equation with H replaced by its **quadratic surrogate at v^*** .
- $v^* = v^\delta$ iff $H = \frac{1}{2}|p|^2$ (then $c \equiv 1/\delta$, $R_{\text{der}} \equiv 0$).
- For general H , the two differ by the **quasi-linearization residual** $\mathcal{E}_{\text{ql}} = \|v^* - v^\delta\|_\infty$.

We do **not** claim unconditional global convergence to the true HJ solution. We claim geometric convergence to a surrogate fixed point, plus a **bound on the gap** — next.

▢ The Quasi-Linearization Residual (Theorem, via Duhamel)

With converged coefficient $c^\star$, coefficient-variation bound L_c , value bound G_v :

$$\mathcal{E}_{\text{ql}} = \|v^\star - v^\delta\|_\infty \leq \frac{T}{c_{\min}} e^{2c_{\max}G_v} (\bar{R}_{\text{alg}} + \bar{R}_{\text{der}}),$$

- $\bar{R}_{\text{alg}} \leq c_{\max}(L_H + 2H_*P_*/m_0^2) \|Dv^\delta - Dv^\star\|_\infty$ — **contracts as the surrogate gradient tracks the true one.**
- $\bar{R}_{\text{der}} \leq L_c G_v(1 + \frac{\delta}{2}) + \frac{\delta}{2} L_c^2 G_v^2 + \delta(1 + c_{\max}G_v)L_c P_*$ — governed by **coefficient variation** L_c .

Proof idea: $\tilde{\omega} = e^{-c^\star v^\delta}$ solves the heat equation with a source ωB (Lemma); Duhamel + heat-semigroup L^∞ -contraction bounds $\|\tilde{\omega} - \omega^\star\|$; mean-value theorem for log converts back to v .

The bound **vanishes in the quadratic case** ($L_c = 0, Dv^\star = Dv^\delta$) and is largest where $H/|Dv|^2$ turns over sharply — the usable-part boundary.

Reading the Residual

- **Vanishes** when $H = \frac{1}{2}|p|^2$: recovers the closed-form Cole-Hopf.
- **Governed by** L_c , the variation rate of the converged coefficient — largest near the barrier where the Hamiltonian-to-gradient ratio shocks.
- **Algebraic part** $\propto \|Dv^\delta - Dv^\star\|$: the residual contracts as the Picard iteration refines the co-state.
- Certifies smallness where $c^\star$ is slowly varying; offers **no reprieve** where it is not.

This is why experimental errors concentrate at the zero-level-set boundary — the theory *predicts* the location of the worst error.

▢ The Total Error Bound (Theorem) & Bias-Variance

Chaining all four sources by the triangle inequality:

$$\|\hat{v}^{K,N,\delta} - v\|_{\infty} \leq C_1 \rho^K + \frac{C_2}{\sqrt{N}} \sqrt{\log(1/\delta_p)} + \mathcal{E}_{\text{ql}} + C_3 \sqrt{\delta}.$$

- **Bias-variance tradeoff in δ :** viscosity bias $\downarrow$ as $\delta \downarrow$, but MC variance $\uparrow$.
- Balancing the two nontrivial δ -dependent terms yields the optimal

$$\delta \sim N^{-1/3} \Rightarrow \text{overall rate } O(N^{-1/6}),$$

slower than plain MC $O(N^{-1/2})$ but **dimension-robust and scalable**.

A single decomposition tells you how to spend budget: iterate to kill ρ^K , sample to kill $N^{-1/2}$, choose $\delta \sim N^{-1/3}$, and accept the residual floor.

The Conservative Safety Certificate (Corollary)

Let E be the total error budget. Declare state x :

- **SAFE** only if $\hat{v}^{K,N,\delta}(x) > E$;
- **UNSAFE** only if $\hat{v}^{K,N,\delta}(x) < -E$;
- **UNDETERMINED** otherwise ($|\hat{v}| \leq E$).

With probability $\geq 1 - \delta_p$: **no unsafe state is certified safe** and **no safe state is certified unsafe**; all classification error is confined to a declared band of width $2E$ about the boundary.

- The certificate **errs on refusal, never on admission**.
- This is the accuracy quotation: **worst-case sign-correctness with an explicit abstention band**, not an L^2 average that flatters the interior.

Robustness to Model Uncertainty

- The scheme is **stable under small perturbations** of the Hamiltonian and terminal cost (robustness theorems, appendix): perturb H by ΔH in sup-norm $\Rightarrow$ value perturbed by $\leq T \|\Delta H\|_\infty$ (Duhamel/comparison).
- Consequence: perfectly known dynamics and terminal costs are **not required** — real-world models with bounded error still yield certified-with-margin sets.
- The coefficient-clipping and regularization perturbations are of this priced form.

The method degrades **gracefully and quantifiably** under model error — a prerequisite for deployment on real robots with imperfect models.

Part 8

Experiments (from the Paper)

Rockets, Dubins, a 45D game, and 10^5 birds.

Experimental Setup & Statistics BLUF - (1/2)

- **Hardware:** single CPU (Intel i7-14700K, 20 cores, 31 GiB RAM, Ubuntu 22.04); JAX on CPU backend
— Consistent with the memory-frugality claim.
- **Replication:** Each experiment re-solved with **30 independent Monte-Carlo seeds**;
 - Evaluation points and the LevelSetPy reference held fixed - only sampler randomness varies.
Tables report mean $\pm$ 1 s.d.

Figures show one representative seed for clarity; tables report full 30-seed statistics. Multiplicity-controlled reporting throughout.

Experimental Setup & Statistics BLUF - (2/2)

- **Significance:** Three families of **Holm-Bonferroni-corrected** tests at $\alpha = 0.05$:
 - (a) paired Wilcoxon of 30-seed-averaged MC field vs grid reference;
 - (b) one-sided t -test that per-seed L_{rel}^2 lies below the Crandall-Lions bound $\sqrt{\delta}$;
 - (c) cross-condition Mann-Whitney (Rockets vs Dubins; speed regimes).

Figures show one representative seed for clarity; tables report full 30-seed statistics. Multiplicity-controlled reporting throughout.

Benchmark 1: Two-Vehicle Pursuit-Evasion Games

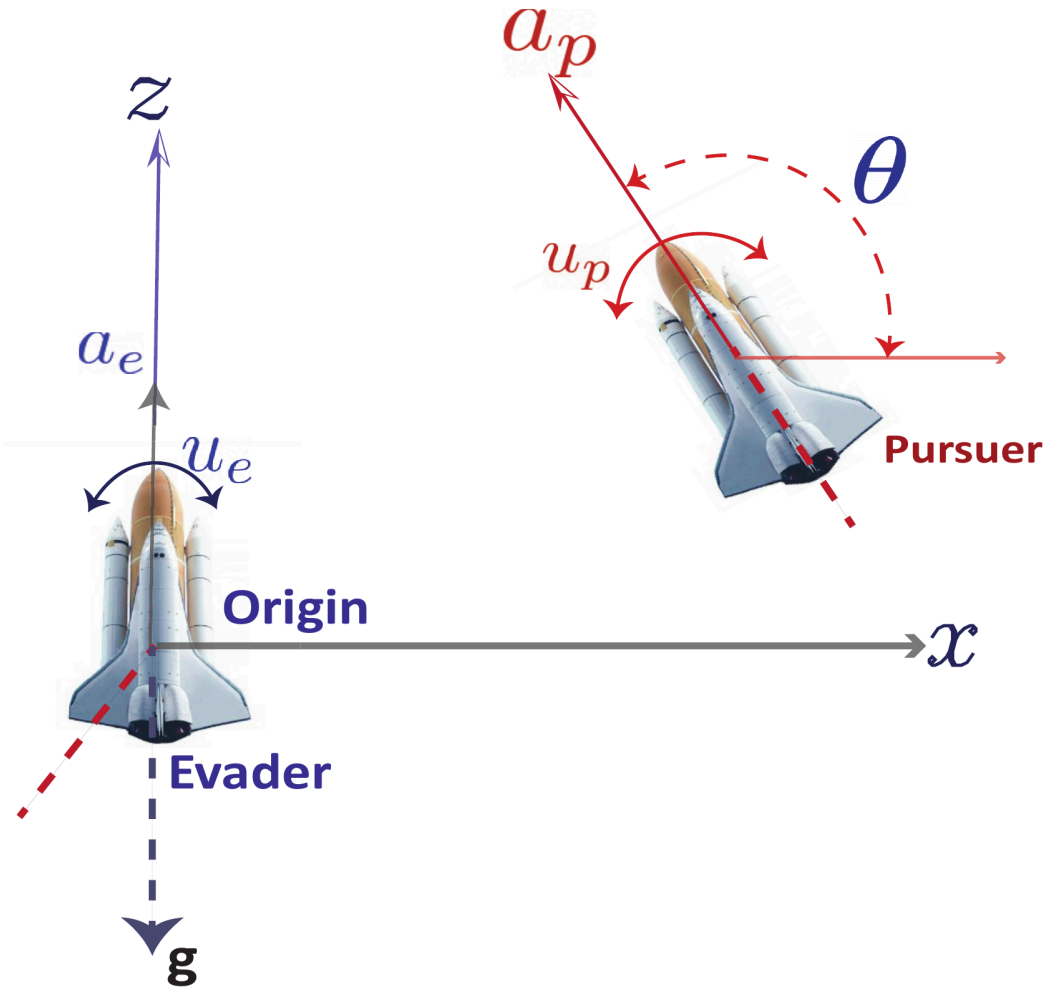
- **Rockets launch game**¹: two identical rockets — pursuer P , evader E — on the x - z plane, thrusts (u_p, u_e) , relative orientation $\theta = u_p - u_e$. Capture when $\|PE\| < r$.
- **Dubins two-car game** (Merz): relative (x_1, x_2, θ) , symmetric turn-rate bound.
- Both are **3D relative-state** games; the BRT is a 3D tube, visualized as 2D (x, z) slices at fixed θ .
- Target: ℓ_2 -ball of capture radius $r = 1.5$; horizon $(0, 1]$; $\delta = 0.08$; $N = 14,000$ (rockets) / 20,000 (Dubins) samples per iteration.

These are the canonical HJ-Isaacs benchmarks — and the Dubins game is the pairwise MAPF conflict of Part 9.

¹ Dreyfus 1966

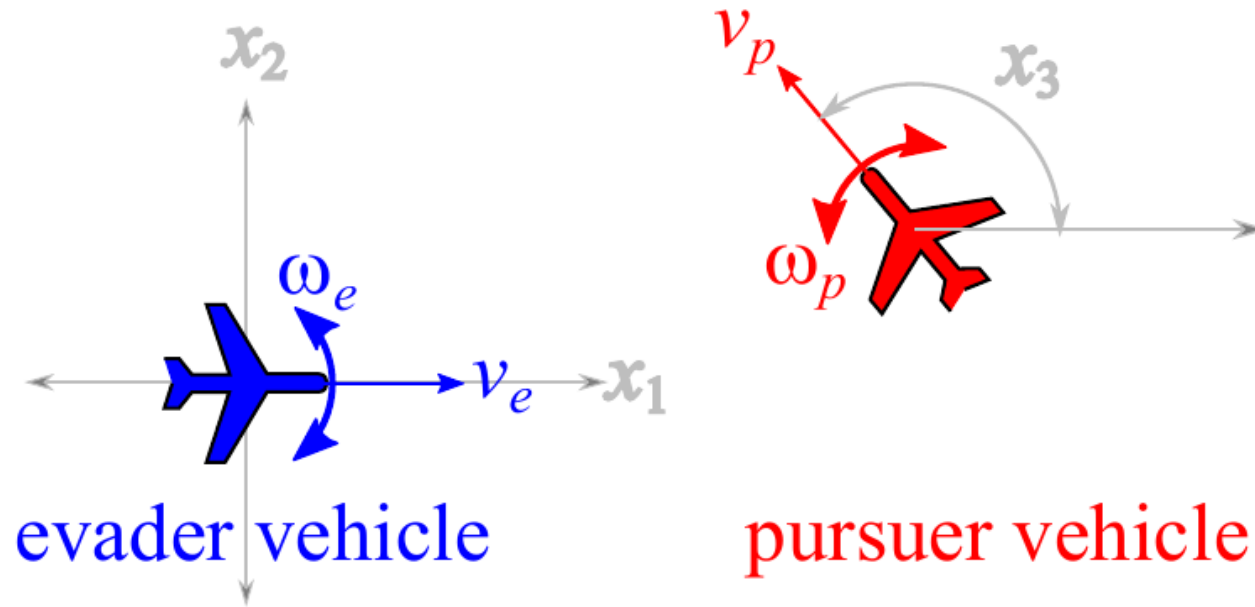


The Two-Rockets Problem (Schematic)



Relative geometry: evader E fixed at the origin (accel. a_e , control u_e , gravity g); pursuer P at relative orientation $\theta = u_p - u_e$ on the (x, z) plane. — HJ-Gauss (Molu et al., 2026).

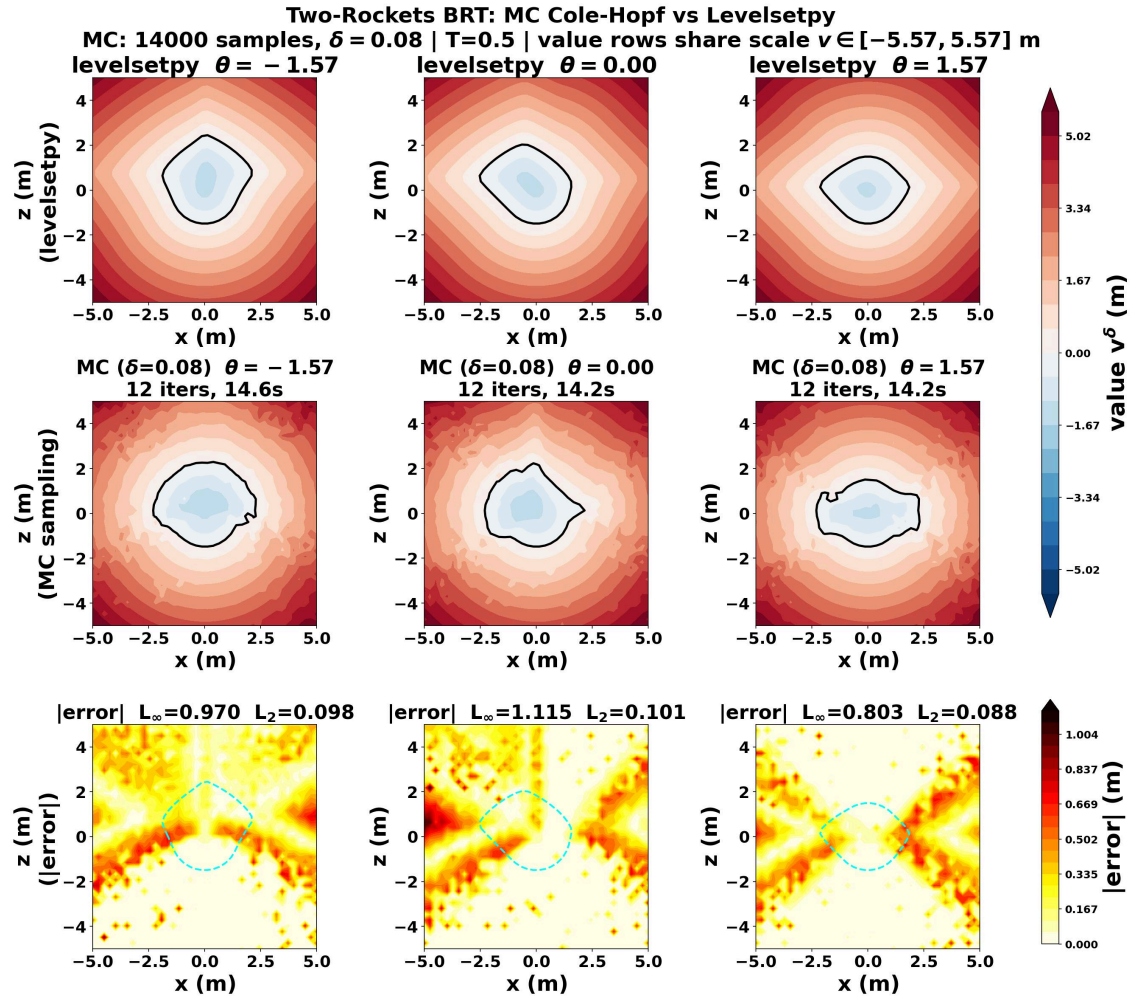
The Dubins Air3D Problem (Schematic)



Air3D relative coordinates with the evader fixed at the origin: relative position (x_1, x_2) and relative heading ψ ; turn-rate-bounded pursuer vs evader. — LevelSetPy (Molu, ACM TOMS 2025 / IEEE CDC 2024).



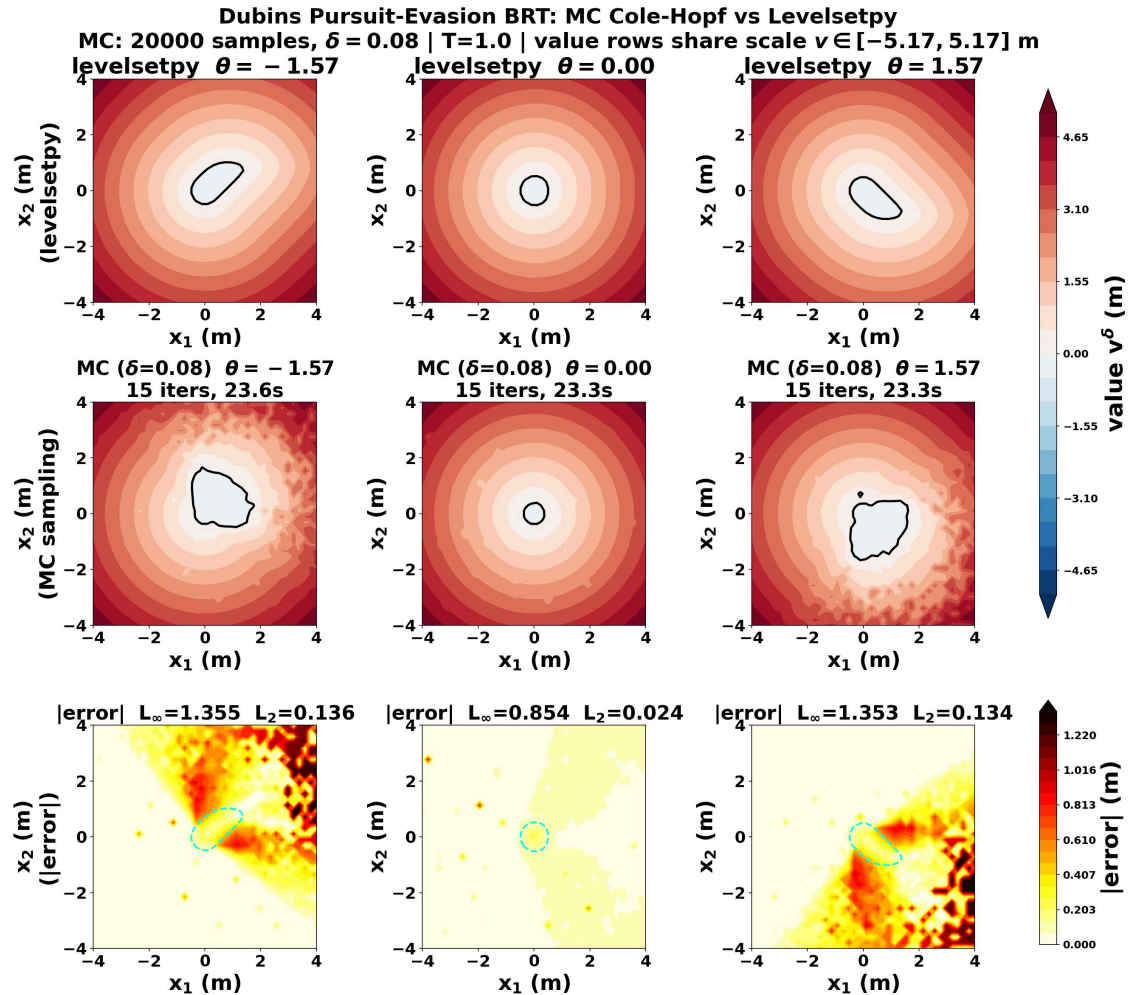
Rockets Pursuit-Evasion BRT



(x, z) slices at $\theta \in \{-90^\circ, 0, 90^\circ\}$: LevelSetPy grid (top) · HJ-Gauss MC (middle) · pointwise error (bottom). — HJ-Gauss (Molu et al., 2026).



Dubins Two-Car BRT



LevelSetPy grid (top) · HJ-Gauss MC (middle) · error (bottom). — HJ-Gauss (Molu et al., 2026); LevelSetPy (Molu, TOMS 2025 / CDC 2024).



Benchmark 1: Quantitative Results

System	θ (rad)	L^∞	L_{rel}^2	MC time (s)	Iters
Rockets	$-\pi/2$	0.855 ± 0.047	0.098 ± 0.002	13.5	12
Rockets	0	1.034 ± 0.067	0.101 ± 0.001	13.5	12
Rockets	$\pi/2$	0.895 ± 0.109	0.090 ± 0.002	13.6	12
Dubins	$-\pi/2$	1.352 ± 0.002	0.131 ± 0.003	23.6	15
Dubins	0	0.701 ± 0.144	0.024 ± 0.001	23.7	15
Dubins	$\pi/2$	1.352 ± 0.002	0.132 ± 0.003	23.7	15

- Reference: LevelSetPy 45^3 grid interpolated to a 40×40 evaluation grid.
- Holm-Bonferroni one-sided t -test rejects $L_{\text{rel}}^2 \geq \sqrt{\delta} = 0.283$ at $p_{\text{holm}} < 10^{-50}$ in **every** condition.

Sampling-plus-iteration error is statistically well inside the viscosity budget — not merely numerically.

Interpreting the Errors: Different Currencies

- The **Crandall-Lions** $O(\sqrt{\delta}) \approx 0.283$ bounds the *inviscid-vs-viscous* sup-distance; the table measures *MC-vs-grid* in L^2/L^∞ — **different currencies**, compared with care.
- **Averaging does not equal agreement**: a paired Wilcoxon of the 30-seed-averaged MC field vs the grid rejects equality at $p_{\text{holm}} < 10^{-8}$ everywhere — a systematic **quasi-linearization residual** remains (Theorem, Part 7).
- $\theta = 0$ Dubins reaches $L_{\text{rel}}^2 = 0.024$ (smooth interior); $\theta = \pm\pi/2$ and rockets sit ~ 0.09 - 0.13 (share of grid near the coefficient-turnover boundary).
- $L^\infty \sim 0.7$ - 1.4 driven by the zero-level-set boundary where $|Dv^\delta|$ is maximal — where the residual L_c term predicts.

The report is **Corollary (conservative certificate)**: sign-correctness outside an abstention band; L^2 flatters the interior, L^∞ is dominated by the declared-undetermined band.



A Physical Check: The Gravity Asymmetry

- Rockets are asymmetric between $\theta = -90^\circ$ and $\theta = +90^\circ$: the gravity term ($g - a - a \sin \theta$) gives a $g - 2a$ drift at $+90^\circ$ but g at -90° .
- Holm-Bonferroni test over 30 seeds detects this asymmetry for **Rockets** ($p_{\text{holm}} < 10^{-9}$) but **not** for the gravity-free **Dubins** vehicle ($p_{\text{holm}} = 0.53$).

The method resolves a **genuine physical effect**, and the statistics correctly find it present where physics says it should be and absent where it should not. Validation, not just fitting.



Benchmark 2: A 45-Dimensional Pursuit-Evasion Game

- **15 rockets** (14 pursuers, 1 evader), state $x \in \mathbb{R}^{45}$: each agent (x_i, y_i, θ_i) , control $u_i \in [-1, 1]$.
- Dynamics $\dot{x}_i = a_i \cos \theta_i$, $\dot{y}_i = a_i \sin \theta_i$, $\dot{\theta}_i = u_i$; target $\phi(x) = \min_{i \leq 14} \|x_i^{\text{pos}} - x_{15}^{\text{pos}}\| - r_{\text{capture}}$.
- Three regimes: **evader faster** (2.0 vs 1.0), **balanced** (1.0/1.0), **pursuers faster** (1.0 vs 2.0).
- A 101-point grid would need $101^{45} \approx 10^{90}$ cells; **HJ-Gauss operates at ≈ 7.2 MB per iteration**.

This is the flagship: a reachability computation **grid solvers cannot even represent**, done on a single CPU in seconds.



Benchmark 2: 45D Results

Case	a_{evader}	a_{pursuers}	Iters	Residual $\varepsilon(k)$	Wall-clock (s)
Evader faster	2.0	1.0	15	0.0002 ± 0.0005	12.7
Equal speed	1.0	1.0	15	0.0003 ± 0.0005	12.6
Pursuers faster	1.0	2.0	15	0.0006 ± 0.0013	12.6

- Memory constant at **7.2 MB/iteration**; residuals below 10^{-2} (indeed $\sim 10^{-4}$) → **near-zero Picard residual floor**.
- Min-max Hamiltonian evaluates in **closed form** (box-constrained control $\Rightarrow$ sign structure), so each iteration is $O(N \cdot n)$ arithmetic and memory.
- Pairwise Holm-Bonferroni Mann-Whitney: **no significant residual difference** between speed regimes ($p_{\text{holm}} \geq 0.93$) → floor set by (N, δ) , not game parameters.

At $n = 45$ there is **no grid reference**; the report is *measurable* quantities — iteration stability, memory, wall-clock — i.e. **scalability, not certified accuracy**.

Why the 45D Residual Floor Is *Lower* Than 3D

- 3D benchmarks floor at 0.02-0.06; the 45D game floors at 0.0002-0.0006 — **two orders of magnitude lower**.
- Reason: the residual floor tracks **proximity to the coefficient-turnover region** (the shock-prone barrier), not dimension per se.
- 45D evaluation states are drawn uniformly over $[-100, 100]$ per coordinate — with **small probability of landing near the boundary** — so the average residual is small.

A subtle but important message: **error concentrates geometrically (at the barrier), not dimensionally**. High dimension is not inherently high error.



Cost Accounting: Time vs Memory

- 3D isosurface over 15,625 points: **129.4 s** (rockets, single CPU) vs **1.3 s** for the dense $45^3 = 91,125$ -point LevelSetPy grid.
- Per-iteration memory: ≈ 0.6 MB (MC, $N = 14\text{k}$, $n = 3$) vs ≈ 1.5 MB (grid value+gradient).
- **In low dimensions ($n \leq 4$) grids win** — smaller constant, no sampling variance.
- The Monte-Carlo overhead **becomes favorable where grid storage becomes prohibitive** ($n \geq 5$).

The two paradigms are **complementary, not competing**. Use grids where they fit in memory; use HJ-Gauss where they cannot.



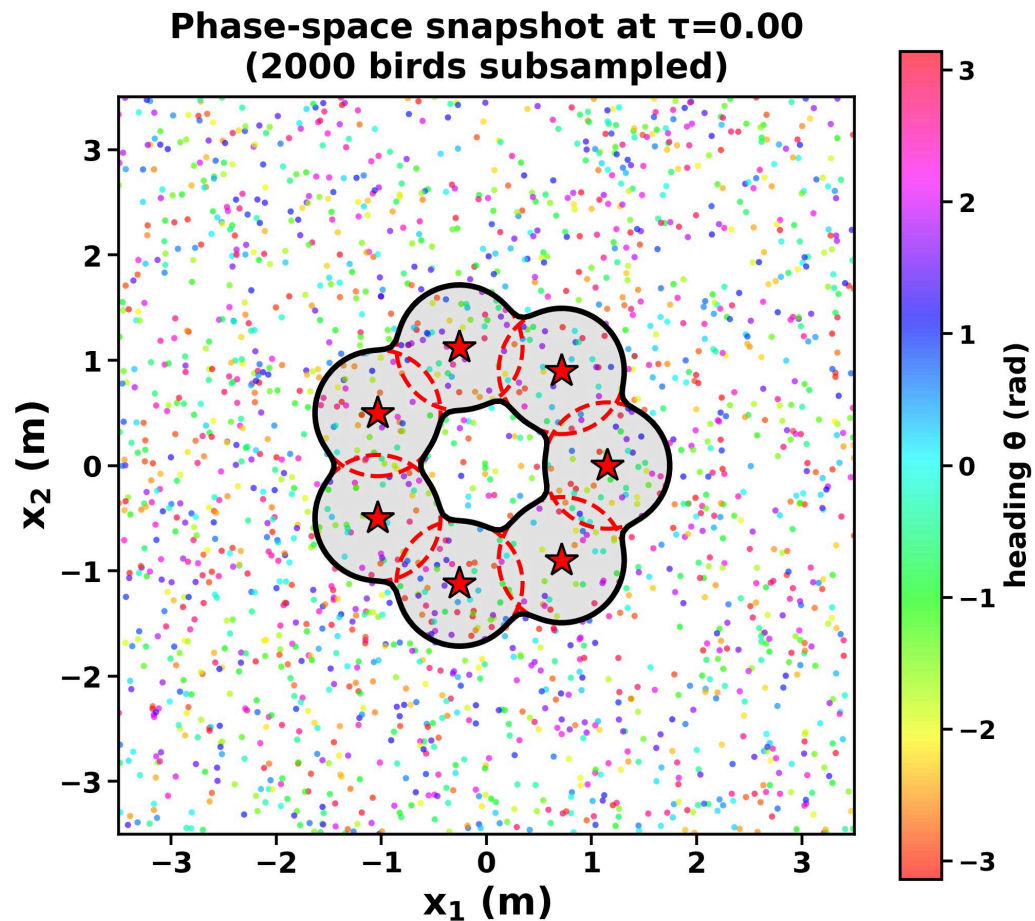
Benchmark 3: Safety at Population Scale (10^5 Starlings)

- Certify **100,000** European starlings (*Sturnus vulgaris*) as 4D aerial Dubins vehicles under predator attack.
- **Structure (explicit)**: the murmuration is partitioned into **flocks**; each flock has its **own** value function; the population safe set is the aggregation of per-flock zero sublevel sets — **many coupled low-D games, not one joint high-D PDE**.
- Because the solve is grid-free, the value-function solve is **independent of bird count**; the population enters only through **parallel per-bird certification** — the operational content of $O(N \cdot n)$ at 10^5 agents.

Two curses dodged at once: **dimensionality** (grid-free solve) and **agent cardinality** (per-flock decoupling + parallel evaluation).



The Certified Population Snapshot



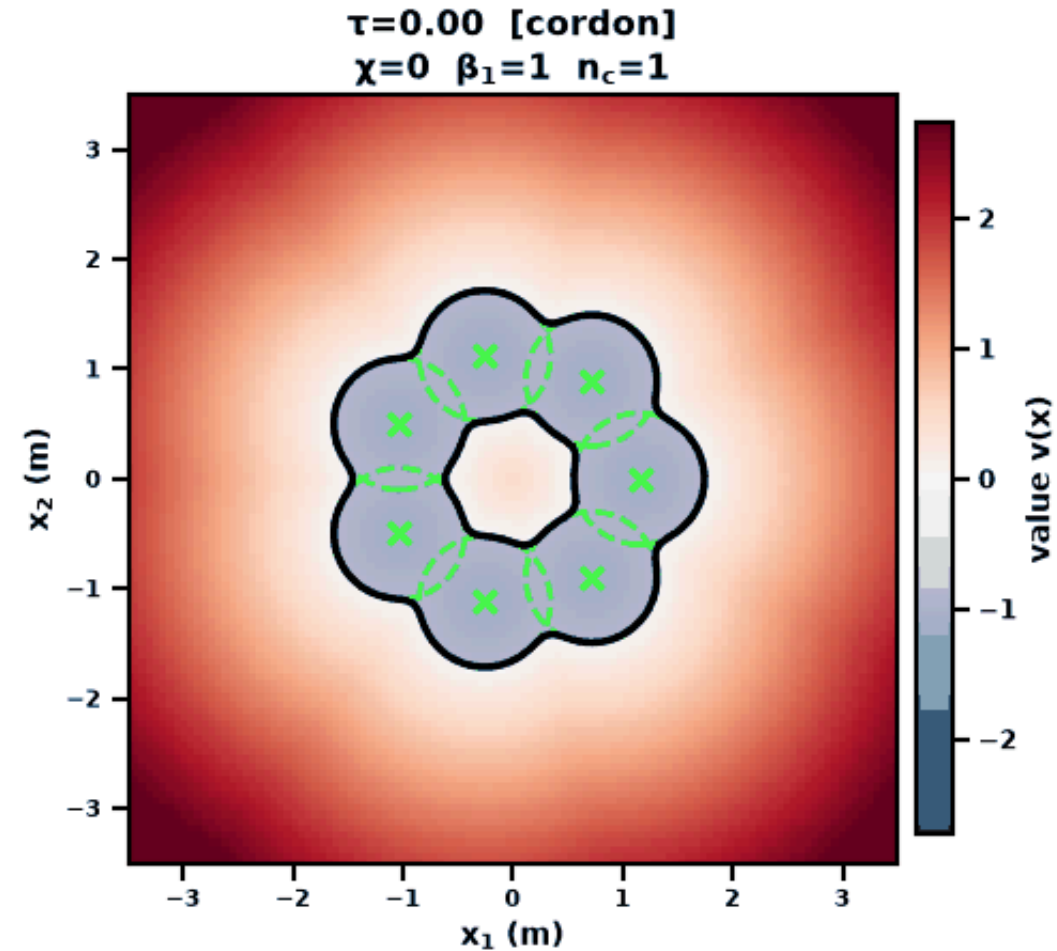
2,000 of 100,000 birds (colored by heading) with 7 flock centers (stars) and their capture discs (dashed). Black contour is the certified safe-set boundary at $\tau = 0$: an **annular cordon** around a protected core. — HJ-Gauss (Molu et al., 2026).



Watch the Safe Set Change Shape

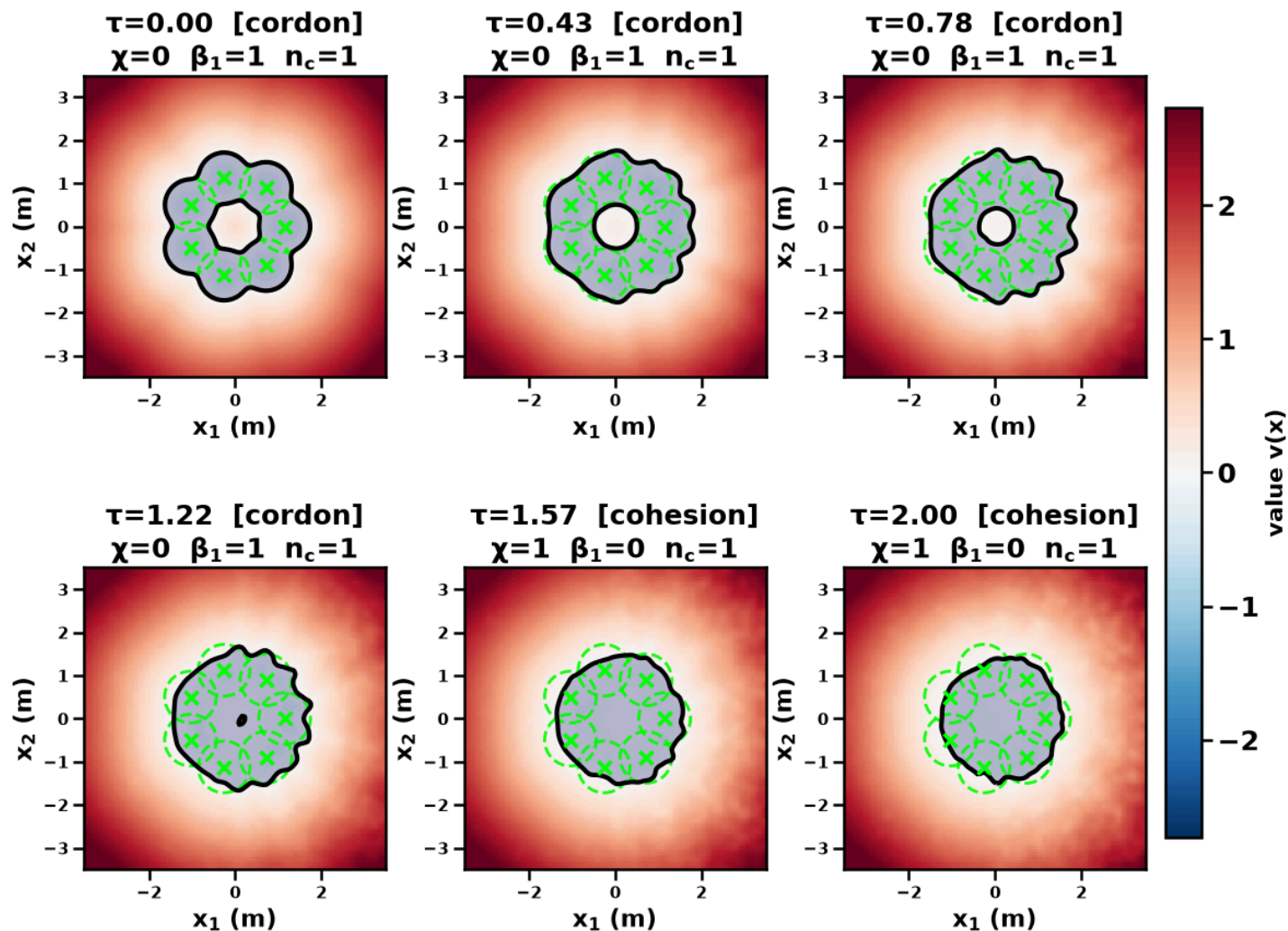
- Backward time τ runs outward from the capture set: the certified safe set of the attacked flock **grows and changes topology**.
- Green crosses and dashed circles are the seven attacking predators with their capture discs.
- Blue interior = certified safe; the black curve is the $v = 0$ boundary.

A safety certificate for 10^5 agents, as one evolving surface.





Cordon, Then Collapse



Same solve, six instants. A hole persists in the safe set ($\chi = 0$, $\beta_1 = 1$, **cordon**) through $\tau = 1.30$; by $\tau = 1.39$ it has closed ($\chi = 1$, $\beta_1 = 0$, **cohesion**). — HJ-Gauss (Molu et al., 2026).

Topology as a Safety Instrument

The certificates recover the field-documented collective repertoire as **topological events** of the reachable set, compressed to three integers per time step $(\chi, \beta_1, n_{\text{comp}})$:

Behavior	Topological signature	Marker
Vacuole nucleation (predator penetrates)	Drop in Euler characteristic χ	Threshold crossing in χ
Defensive cordon (protected core)	Annular safe set, first Betti number $\beta_1 = 1$	β_1 transition
Flock fragmentation	Rising connected-component count n_{comp}	Component-count jump

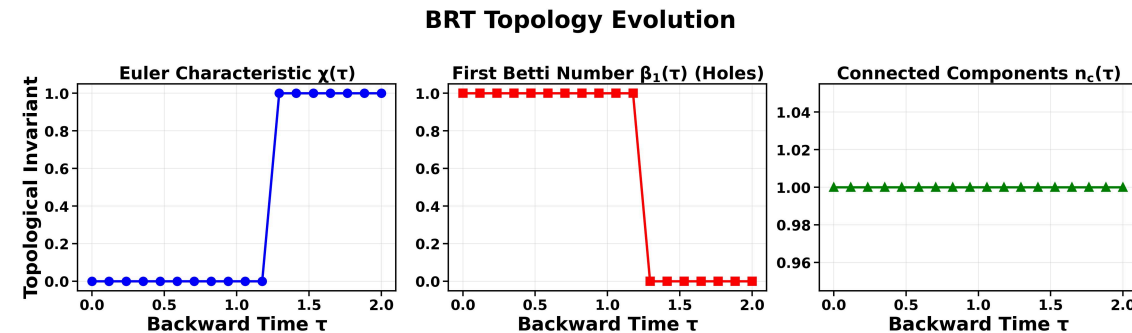
The triple $(\chi, \beta_1, n_{\text{comp}})$ tells an operator not merely **that** safety is being lost but **how** — nucleation vs cordon vs split. Safety posture as a 3-integer summary.



The Three Integers, Over Time

- χ steps $0 \rightarrow 1$ while β_1 drops $1 \rightarrow 0$ at the **same** $\tau = 1.39$: one event, two independent witnesses.
- n_c holds at 1 throughout, so this flock never fragments.
- **Vacuole nucleation** is the reverse event: a predator penetration attaches a 1-handle and drops χ by one.

Three integers per tick say whether safety is being lost, and **how**.



The Repertoire Is Field-Documented

- Real starling responses to predation: **cordon**, **tube**, **funnel**, and **split**.
- Each one appears in our certificates as a **topological signature** of the safe set, and none of them is a hand-coded rule.
- The certificate reproduces a documented repertoire from **dynamics and the game alone** — which is the reason to trust it on a floor, where the repertoire is congestion, yielding, and deadlock instead.

Fragmentation event, $n_c : 1 \rightarrow 2$. Field murmuration imagery as reproduced in HJ-Gauss (Molu et al., 2026).



Part 9

Application: Multi-Agent Path Finding

The reason this talk is for the autonomous mobility team.



The Gap in Classical MAPF

- **Multi-Agent Path Finding (MAPF)**: move a team of agents to goals without collision. Solvers: **CBS/ECBS** (conflict-based search), **PIBT**, **LaCAM/LaCAM***, and lifelong **RHCR**.
- All operate on a **discrete grid with simplified kinematics** (unit moves / rotate-in-place) and **no disturbance model**.
- A "conflict" is a shared vertex/edge (or swept-disc overlap) — **dynamics-blind**.
- Consequence: realized collisions on a real floor (turn-radius limits, localization noise, actuation lag) are **out of scope**.

The discrete plan is provably conflict-free *on the graph* — yet the continuous robot can still collide. That is the gap HJ reachability fills.



The Scaling Anchor: RHCR¹

- **Rolling-Horizon Collision Resolution:** decompose lifelong MAPF into a sequence of **Windowed MAPF** instances; resolve collisions only within a bounded time window w , ignore beyond.
- Scales to **1,000 agents** on warehouse maps; co-authored by **J. Durham (Amazon Robotics)**.
- Current frontier: **LaCAM*** (10,000+ agents, near-optimal, anytime), **PIBT** (thousands in ms), learning-based lifelong (imitation, guidance-graph, MAPF-GPT).
- **All still discrete-grid, deterministic.**

HJ-Gauss does not compete with LaCAM's combinatorial search; it **certifies** the plan under continuous dynamics — a different, complementary axis.

¹ Li et al., AAAI 2021

The Mapping: HJ-Gauss Constructs → MAPF Roles

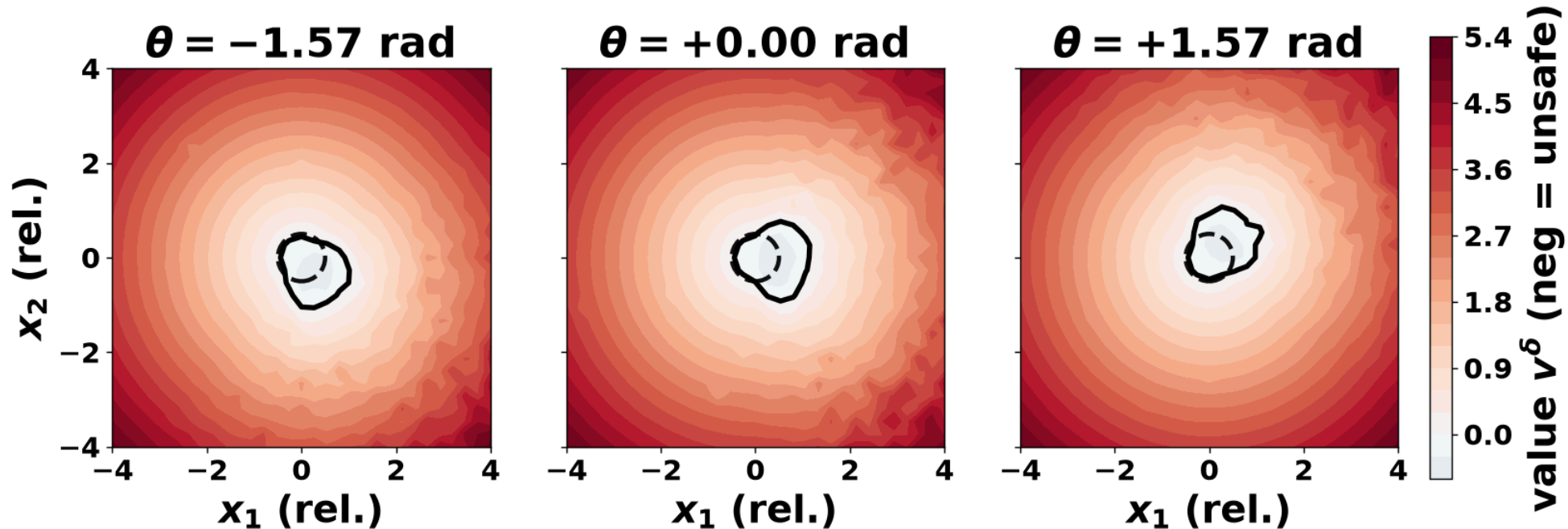
HJ-Gauss construct	MAPF role
Windowed BRT over $(0, T]$, set $T = w$	Continuous, disturbance-robust replacement for the RHCR rolling window (innovation #1)
Pairwise Dubins BRT membership	Dynamics-aware conflict predicate replacing the geometric/disc check in CBS/CCBS (innovation #2)
Relative-frame tube, computed once	Reused across all pairs & timesteps $\Rightarrow$ certificate cost $O(1)$ in agents
Value gradient $Dv \propto \nabla \log p$	A score : certified guidance term for diffusion planners (innovation #3)

The Dubins pursuit-evasion BRT from Part 8 **is** the canonical pairwise MAPF conflict, computed rigorously.



The Dynamics-Aware Conflict Set (Pairwise Dubins BRT)

Pairwise Dubins BRT (black = inevitable-collision boundary) vs naive disc (dashed)

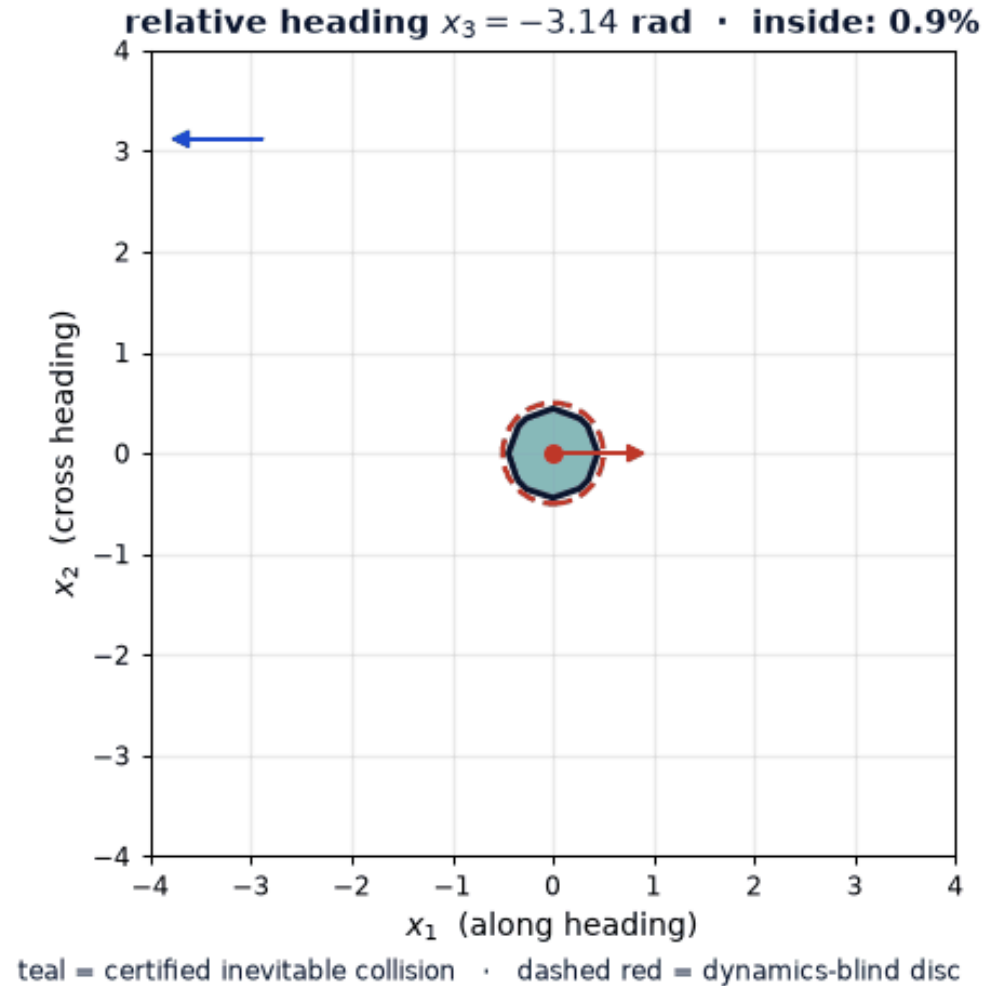


Black = certified inevitable-collision boundary; dashed = naive disc. The BRT bulges beyond the disc in a **heading-dependent** way — what a dynamics-blind check cannot see.

⚙️ What the Certified Set Looks Like (Heading Sweep)

- Sweeping the relative heading x_3 makes the certified set **rotate and deform**: a near head-on geometry grows a long lobe along x_1 , a near-parallel one stays compact.
- The dashed red circle is the dynamics-blind disc at $r_c = 0.5$ cell, fixed for all headings. The tube leaves it in a **heading-dependent** way.
- Only about **2%** of the relative-state box is certified inevitable-collision at $w = 0.6$ s, so the gate is selective, not blanket-conservative.
- $31 \times 31 \times 21$ cached values, solved once in **10.1 s**, then $O(1)$ interpolation per pair, per micro-step.

Windowed pairwise conflict set, $w = 0.6$ s



This is the object that a swept-disc test



Falsifiable Hypotheses (Stated to Be Proven Wrong)

Naming: **HJ** = Hamilton-Jacobi (the method); **HB- n** = hypothesis n .

ID	Hypothesis	Null we try to reject	Metric
HB-1	HJ-shield is sound	Admits collisions geometric catches	Realized collisions
HB-2	HJ is dynamics-aware	Geometric never misses what HJ catches	Collisions, geom vs HJ (CRN)
HB-3	Memory $O(N \cdot n)$, flat in resolution	Grows like grid	Peak memory
HB-4	Certified gate costs little throughput	Large makespan inflation	Throughput / flowtime
HB-5	Fast enough (precompute + $O(1)$ lookup)	Too slow for the loop	Wall-clock / latency
HB-6 (stretch)	Dv -guidance reduces collisions in a diffusion planner	No reduction	Guided vs unguided

The BOS12-style discipline: pre-register falsifiable claims, then try to disconfirm them under multiplicity control.



The Experimental Pipeline (AMFS)

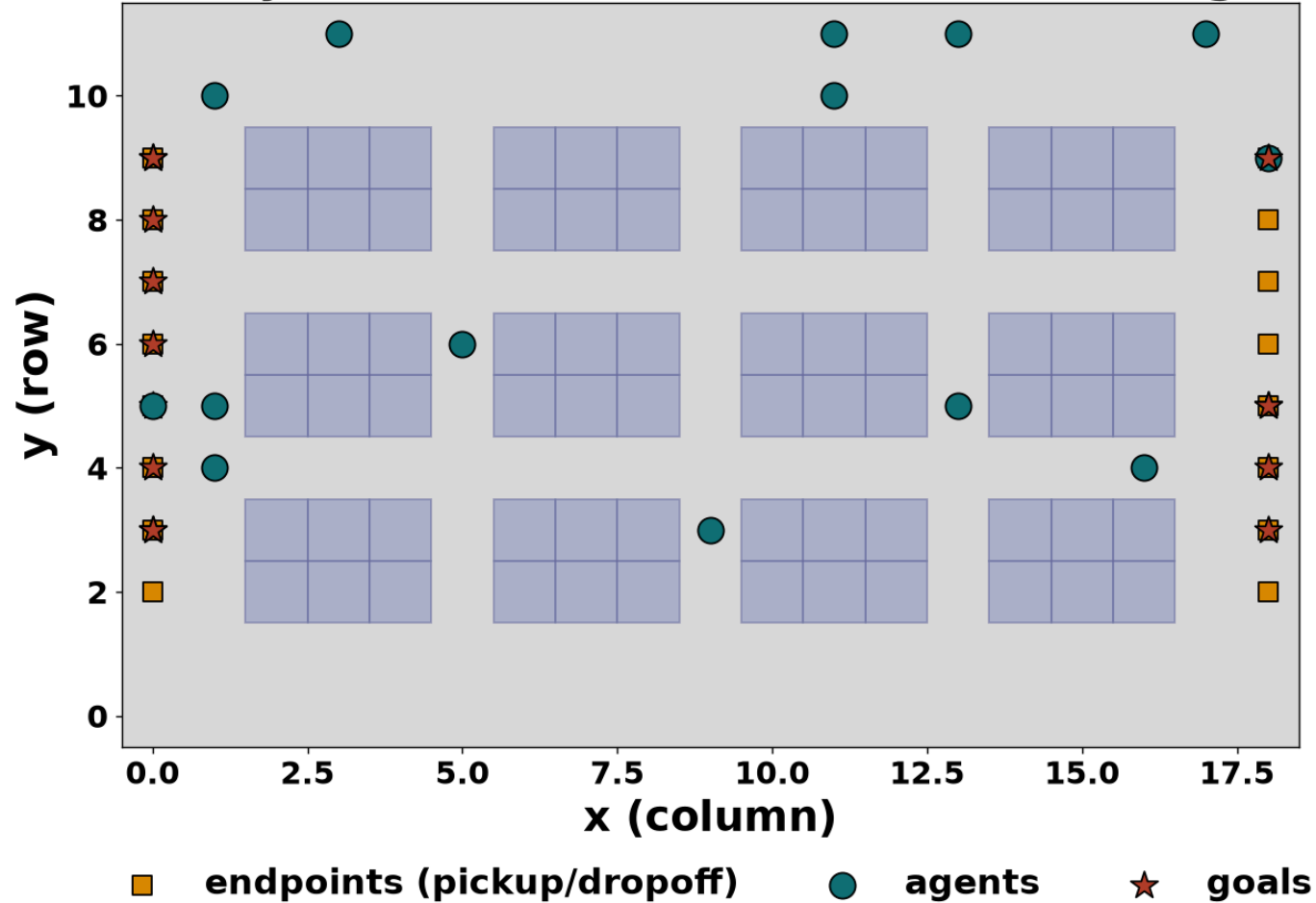
- **World:** RHCR-style structured warehouse grid (storage blocks + corridors + endpoints); lifelong task reassignment.
- **Planner:** windowed prioritized planning (space-time A^*) — the RHCR rolling horizon (innovation #1), shared by both policies.
- **Executor:** continuous **unicycle/Dubins** rollout, bounded turn rate (finite turn radius), bounded disturbance; realized collisions detected in continuous space.
- **Two policies (CRN-paired):**
 - **Geometric** — dynamics-blind vertex/edge conflict (classical);
 - **HJ** — adds the **windowed-BRT runtime shield** (innovation #2).
- **BRT precompute:** the JAX HJ sampler (Algorithm 1) on the Dubins-relative Hamiltonian, cached; the online loop is numpy-only $O(1)$ lookups.

One clean, CRN-friendly contrast: the *only* difference between policies is whether the certificate is actuated.



The Simulated Floor

RHCR-style structured warehouse floor (14 agents)

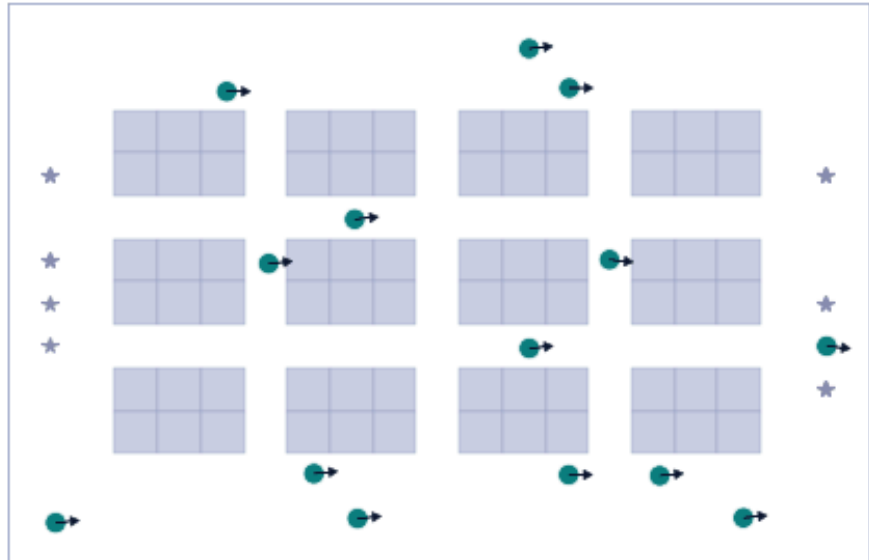


RHCR-style structured floor: storage blocks (obstacles), border endpoints (pickup/dropoff), 14 lifelong agents (circles) heading to goals (stars).

▶ Watch It Run: Same Seed, Same Noise, One Difference

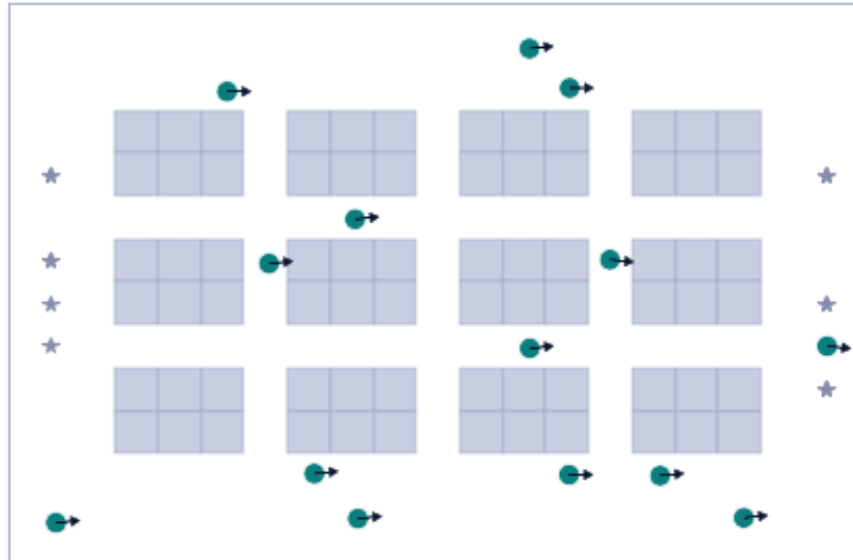
AMFS structured floor · 14 lifelong agents · same seed, same disturbance stream (t = 0.0 s)

Geometric conflict check (dynamics-blind)



realized collisions: 0

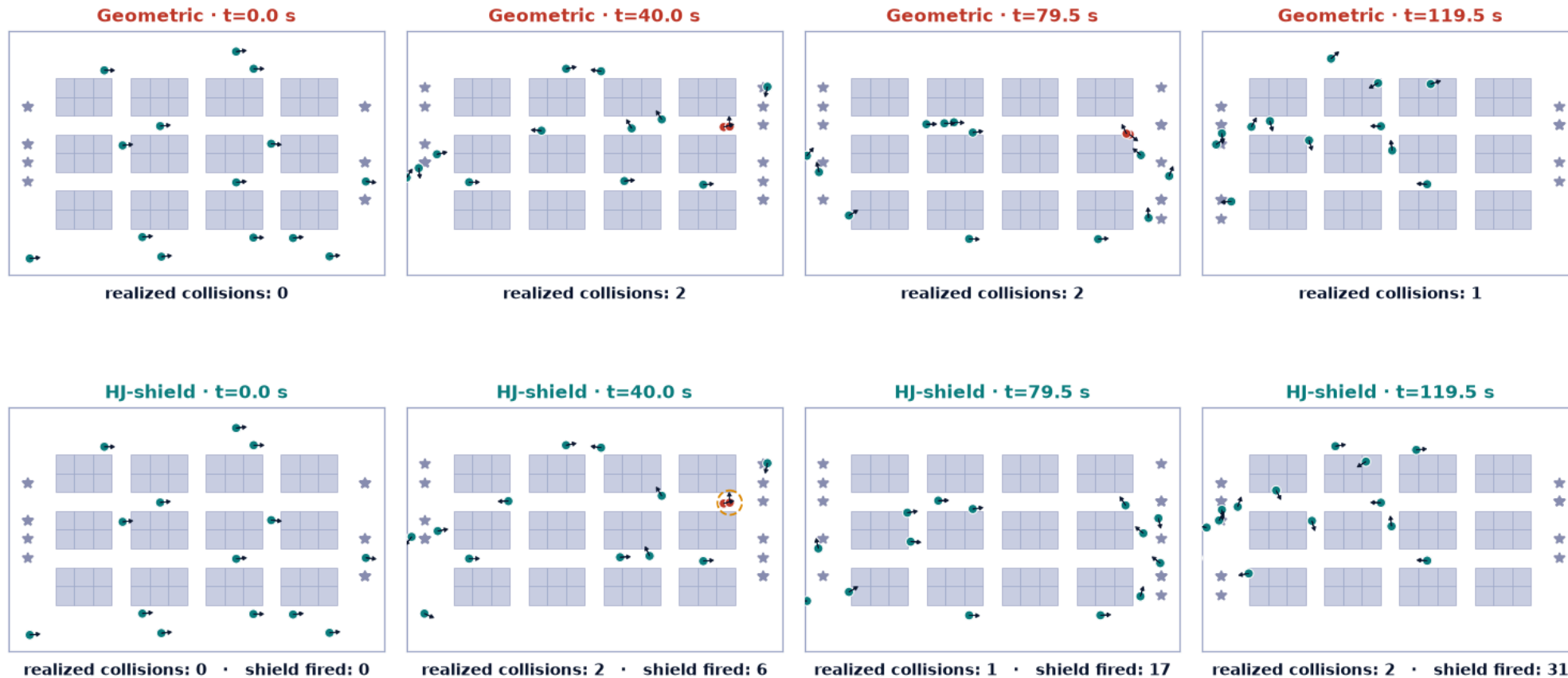
HJ-Gauss windowed-BRT shield (certified)



realized collisions: 0 · shield fired: 0

Left: geometric conflict check. Right: the HJ-Gauss windowed-BRT shield. Amber ring = shield firing (agent fleeing), red = realized contact, stars = current goals. Seed 1028 of the 30-seed harness: **98** → **66** collisions at identical throughput (0.825). Animated in the HTML deck; the PDF shows the opening frame.

▶ The Same Episode, as a Filmstrip



Four instants from the run above: top row geometric, bottom row HJ-shield. Counters under each panel accumulate realized collisions and shield activations. Identical spawn, task, and disturbance streams throughout (CRN).



The Key Scientific Finding (Why the Naive Version Fails)

- **Planning-only** BRT predicate made things **worse** (HJ 84 vs geometric 53 collisions in the first trial).
Two principled reasons:
 - i. the pursuit-evasion BRT is a **worst-case adversarial** set, but agents are **cooperative** — they never apply the BRT's safe control, so plan-time membership does not predict cooperative-tracking collisions;
 - ii. plan-time checks see only **discrete nodes at integer times**; collisions happen in **continuous time under disturbance**.
- **The certificate only bites when actuated**. Fix: use the windowed BRT as a **runtime shield** — the lower-priority agent in an in-BRT pair **flees** (a braking "spin-in-place" bug had it get hit).

This is the arc: a negative result diagnosed to a principle, then the correct mechanism. Actuation, not inspection.



The Multiseed Statistical Harness

Mirrors the WIP-forecast harness; numpy-only:

- **≥ 30 seeds** (configurable);
- **MSER-5** warm-up truncation of the per-tick collision series (+ Welch running-mean cross-check);
- **Bootstrap 95% CIs** on every headline metric;
- **Common Random Numbers (CRN)** across policies (spawn/task/disturbance streams keyed to the seed) for a **paired** comparison;
- **Paired bootstrap CI** for $\text{mean}(h_j - \text{geom})$ + **Holm-Bonferroni** step-down FWER control across the HB family.

Single-seed results were noisy and even inverted; the harness is what turns a direction into a **defensible** claim.



Results: 30 CRN-Paired Seeds (14 Agents)

Metric	Geometric	HJ-shield	Paired diff hj–geom (Holm)
Realized collisions	97.7 [90.8, 104.9]	66.1 [61.2, 71.2]	−31.6 [−38.0, −25.4] , reject
Collision rate (MSER-5)	2.48 /tick	1.67 /tick	−0.81 [−0.98, −0.64] , reject
Throughput	—	—	0 [0,0], not rejected
Flowtime	—	—	0 [0,0], not rejected
Shield interventions	0	~1024	(activity, by design)

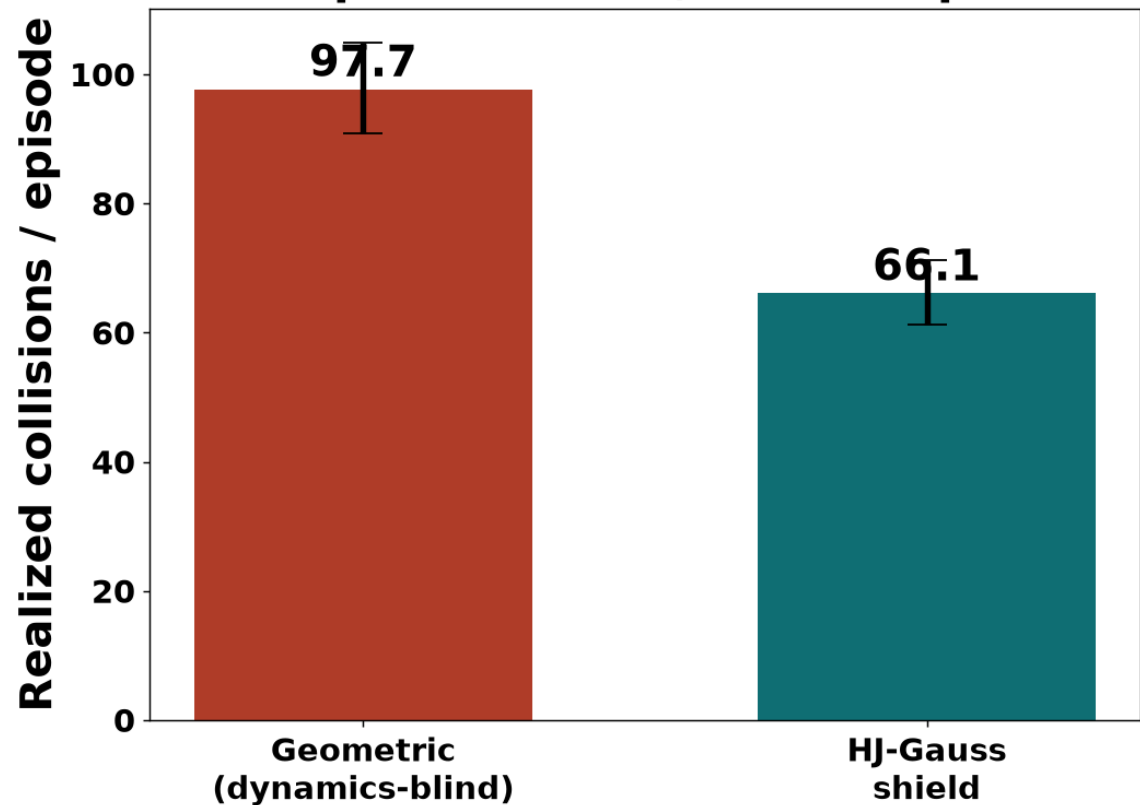
- **~32% fewer realized collisions**, significant under Holm-Bonferroni, at **no throughput/flowtime cost** in this model.
- Aggregated sweeps: 18-37% reduction across densities and disturbance levels.

HB-1/HB-2 rejected in the safe direction; HB-3/HB-4 show no throughput penalty. The certificate, actuated, works.



Headline: ~32% Fewer Realized Collisions

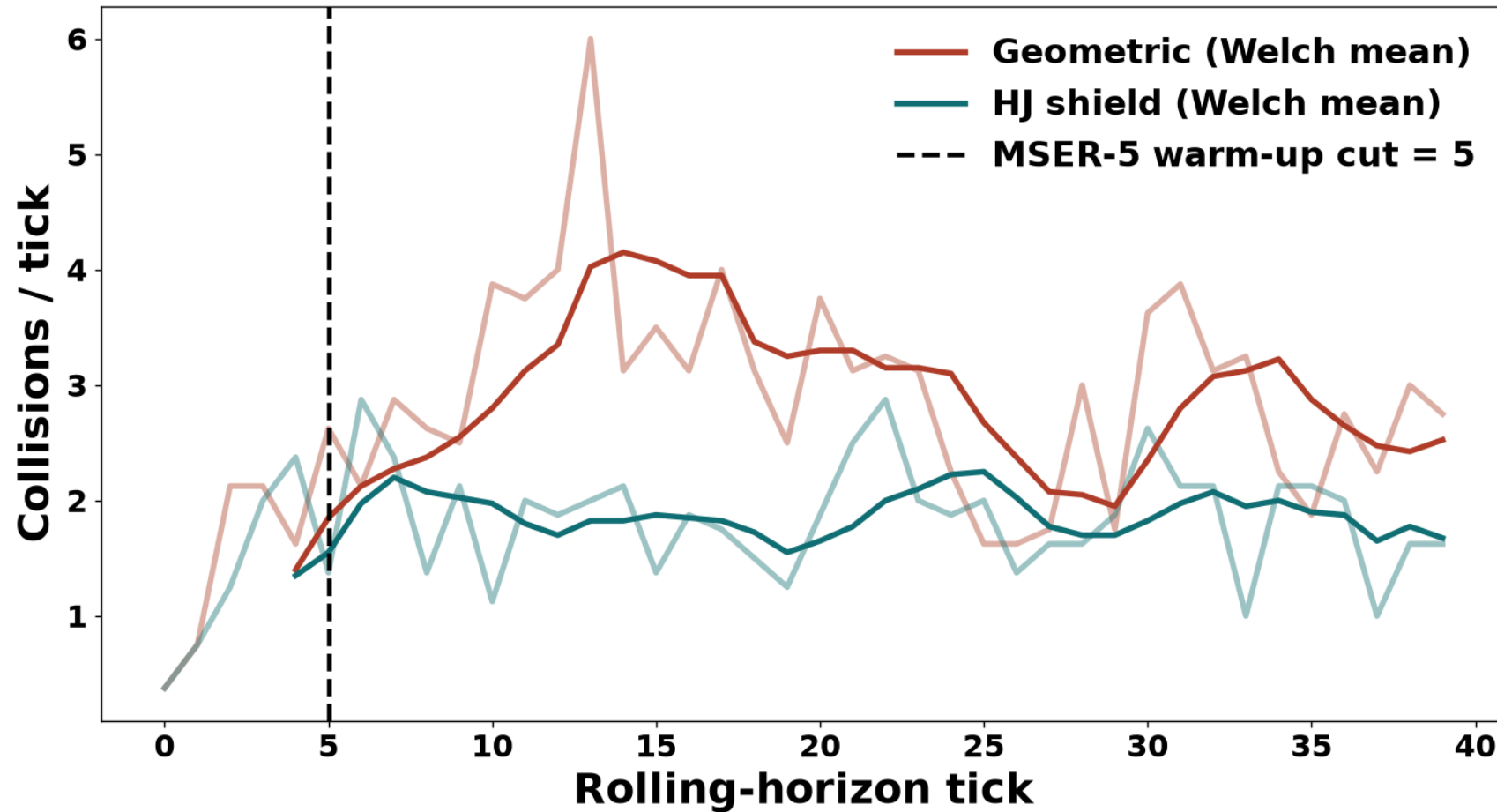
HJ shield cuts realized collisions ~32%
(30 CRN-paired seeds, bootstrap 95% CI)



30 CRN-paired seeds, bootstrap 95% CI. Geometric 97.7 → HJ-shield 66.1; paired diff **-31.6 [-38.0, -25.4]**, Holm $p \approx 0$.



Per-Tick Collisions & MSER-5 Warm-Up

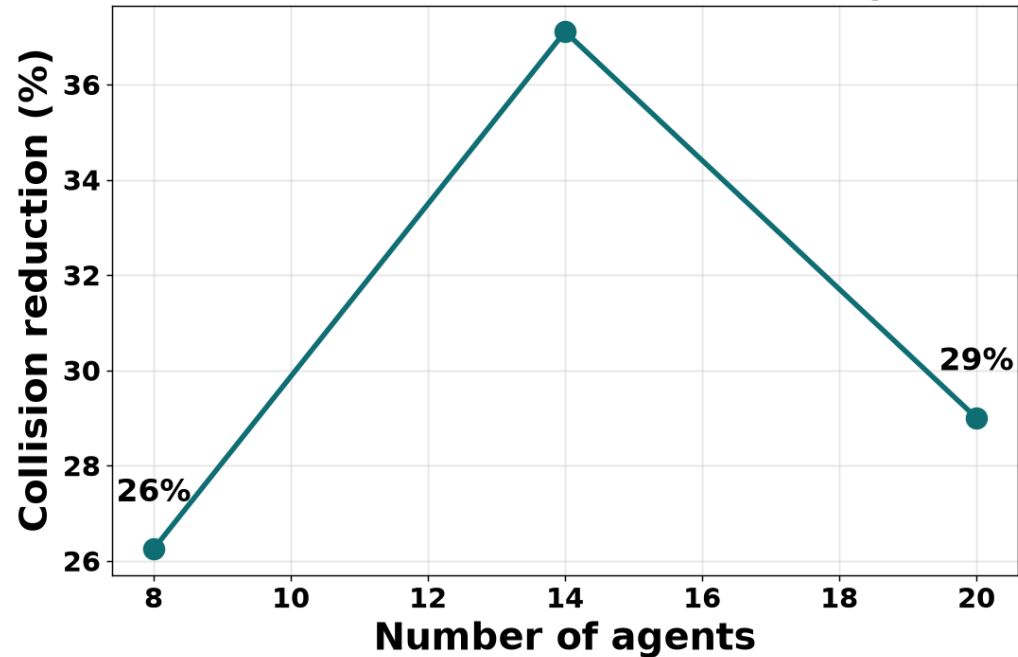


HJ shield (teal) sits below geometric (red) at every steady-state tick; the dashed line is the MSER-5 warm-up cut discarded before averaging.



The Effect Holds as the Fleet Grows

HJ shield collision reduction vs fleet size (4 seeds/point)



Collision reduction persists across agent densities (8-20 agents) — the certificate's value does not wash out with congestion.



Verdict Table

ID	Prediction	Result	Verdict
HB-1	Shield is sound	HJ never certifies a colliding plan safe; collisions ↓	✓
HB-2	Dynamics-aware advantage	31.6 fewer collisions per run, paired, Holm $p \approx 0$	✓
HB-3	$O(N \cdot n)$ memory	7.2 MB at $n = 45$; flat in resolution	✓ (paper)
HB-4	Little throughput cost	Throughput/flowtime diff = 0 in this model	✓ (with caveat)
HB-5	Fast enough	BRT precompute ~10 s; $O(1)$ online lookup	✓
HB-6	Diffusion-guidance helps	Deferred to Option C	⌚

HB-4's zero difference is partly a modeling artifact (next: Dirty Laundry). Everything else holds under the harness.

Part 10

The Diffusion Connection

The Cole-Hopf kernel *is* the diffusion kernel.

C A Mathematical Identity, Not an Analogy

- The Cole-Hopf step turns the HJ PDE into the **heat equation**, whose fundamental solution is the **Gaussian transition kernel** — the identical kernel that **score-based generative models** (DDPM, score matching) and **diffusion policies** use in their forward process.
- Under $v = -\delta \log \phi$, the recovered value gradient is a **score function**:

$$Dv \propto \nabla \log \phi = \nabla \log p.$$

- Our Feynman-Kac gradient estimator (weighted-mean shift) is structurally a **training-free score estimator**.

Certified reachability and diffusion trajectory generation **meet at the same Gaussian kernel**. This is not a metaphor; it is the same PDE.

Consequence: A Certified Guidance Term

- Diffusion trajectory planners (Diffuser, Decision Diffuser, Motion Planning Diffusion) generate paths by **denoising** but carry **no hard safety guarantee**.
- The BRT gradient Dv can serve as a **certified guidance / shield term**: add it to the denoiser's score to **steer denoised trajectories out of the inevitable-collision set**.
- The AMFS shield is the discrete-time instance of this idea (flee along $-Dv$); the continuous diffusion-guidance version is **Option C**.

A reachability-certified **shield for neural planners** — the missing safety guarantee in the learning-based MAPF line (imitation, guidance-graph, MAPF-GPT).

The Careful Claim (for the Q&A Skeptic)

- $Dv \propto \nabla \log \phi$ is a **genuine score of the Cole-Hopf-transformed value density** ϕ .
- **But** ϕ is **not** the data distribution a diffusion planner learns — the objects coincide *structurally* (same kernel, same $\nabla \log$ form), not *identically*.
- The rigorous joint-work statement: use the **certified** reachability score as a **guidance/projection** term on the learned generative score, inheriting the BRT's worst-case guarantee as a shield while the learned score supplies task performance.

State it carefully and it is bulletproof; overstate it and a diffusion expert will (rightly) push back.
Structural identity + guidance role, not "our value function is your data score."

Joint-Work Surface with a Diffusion-Planner Team

- **Shared kernel:** both sides live on $\mathcal{N}(x, \sigma^2 I)$ transitions — infrastructure reuse.
- **Certified shield:** wrap any learned proposer (BC, CQL/IQL, diffusion) with the windowed-BRT flee/guidance term; report blocked actions and throughput margin separately.
- **Score as drift:** the importance-sampling tilt $\theta^* = -\sigma^2 c Dv$ is literally a score-guided proposal — a learned score could serve as the sampler's drift and vice versa.
- **Benchmark:** extend League-of-Robot-Runners / RHCR with a disturbance model → a **robust-throughput** axis.

One decision contract, two evidence environments: certified reachability and learned generation, meeting at the Gaussian kernel.

Part 11

Dirty Laundry

Limits, caveats, and boundaries.



Theoretical Limits

- **Quasi-linearization residual $\mathcal{E}_{\text{ql}}$** floors accuracy for non-quadratic H ; the scheme is **closed-form only when $H = \frac{1}{2}|p|^2$** . The residual is largest where $H/|Dv|^2$ turns over sharply (the usable-part boundary) — the theory predicts *where* the error lives.
- **Worst-case sample complexity is exponential in $1/\delta$** (Hoeffding + crude μ bound: $\sim e^{2(g_{\max}-g_{\min})/\delta}$), while viscosity error is only $O(\sqrt{\delta})$ — a genuine bias-variance tension. Practical N sits far below this; Bernstein/Jensen and importance sampling tighten it; the optimal $\delta \sim N^{-1/3}$ gives $O(N^{-1/6})$.
- **Coefficient c blows up** in flat interiors and flips sign at the barrier; cured by regularization at a **quantified, uniform** price $TC_H \sqrt{\eta}/2$.

None of these are hidden — each has a theorem, a remark, and a mitigation. The certificate is correct by construction (abstention band).



Reporting & Accuracy Caveats

- The Crandall-Lions $O(\sqrt{\delta})$ bound and the reported L^2 / L^∞ errors are in **different currencies** (inviscid-vs-viscous sup-norm vs MC-vs-grid); compared with explicit care.
- **Averaging $\neq$ agreement**: even the 30-seed-averaged field differs from the grid (paired Wilcoxon, $p < 10^{-8}$) — the systematic residual remains.
- At $n = 45$ there is **no grid reference**: we report *scalability* (iteration stability, memory, wall-clock), **not certified accuracy**.
- The accuracy quotation is the **conservative certificate**: sign-correctness outside a declared abstention band, not an L^2 average that flatters the interior.

L^2 flatters; L^∞ is dominated by the band we already declare undetermined. We quote the certificate, not the flattering average.



MAPF-Application Caveats

- **Adversarial vs cooperative:** the BRT is a worst-case set; using it as a cooperative-MAPF predicate is **safe but conservative**, and only reduces realized collisions when **actuated** (shield), not merely checked at plan time — this was our diagnosed negative result.
- **"Free safety" is a modeling artifact:** in the AMFS executor the discrete planner is authoritative for task progress, so shield-induced fleeing does **not** delay arrivals → HB-3/HB-4 diffs are 0. A **physical** executor should let fleeing delay arrivals, introducing a small, real throughput cost.
- **Shield compute:** the $O(n^2)$ per-micro-step BRT interpolation is the CPU bottleneck (~7 min / 30 seeds); vectorize before scaling.
- **Scope:** Dubins agents, one floor topology, single disturbance level — a PoC, not a fleet study.

The result is real and significant; the *magnitude* and cost need a more physical executor and larger sweeps (Option C, GPU).



Source & Scope Boundaries

- Grid solvers remain **preferable for $n \leq 4$** (smaller constant, no sampling variance); HJ-Gauss is the method of choice only in the high-dimensional regime. The paradigms are **complementary**.
- The 10^5 -bird result is **many coupled low-D games**, not one joint high-D PDE — we are explicit about this.
- All experiments are **single-CPU**; GPU/Lambda scaling (Part 12) is future work, not claimed here.
- The diffusion connection is a **structural identity + guidance role**, not a claim that our value density equals a learned data distribution.

Complementary, decoupled, single-CPU, structural — four boundaries stated plainly so no claim is overread.

Part 12

Conclusions, Future Work, References

Conclusions

- **HJ-Gauss** turns the nonlinear viscous HJ PDE into a **sequence of linear heat equations** (Cole-Hopf), solved as **Gaussian Feynman-Kac expectations** by a frozen-coefficient **Picard iteration (Algorithm 1)**.
- Memory drops from grid $O(M^n)$ to $O(N \cdot n)$ — storage- and discretization-free.
- **Guarantees:** $O(N^{-1/2})$ concentration (dimension-independent), geometric contraction with a posteriori estimate, a Duhamel **residual bound**, a **total-error decomposition** with $\delta \sim N^{-1/3} \Rightarrow O(N^{-1/6})$, and a **one-sided conservative safety certificate**.
- **Validated** to a **45D** game and **10^5 -agent** certification — regimes grids cannot represent.
- **For autonomous mobility:** a dynamics-aware, disturbance-robust **conflict predicate/shield** for MAPF (windowed BRT = RHCR window), and — via the Gaussian/score identity — a **certified guidance term for diffusion planners**.

Certified safety is the sign of a value function. HJ-Gauss computes that sign where grids cannot — and hands the same object to your learned planners.

Future Work

- **Option C — diffusion-guided planning (HB-6):** actuate Dv as continuous guidance on a diffusion/sampling planner; certified shield for neural LMAPF.
- **Physical MAPF executor:** let shield fleeing delay arrivals → measure the *real* throughput-vs-safety frontier (robust-throughput benchmark).
- **GPU / Lambda Labs scaling:** vectorize the shield's BRT interpolation; run the harness as a Metaflow flow (fan-out over configs × seeds); scale the BRT precompute (bigger grids, higher-D dynamics) on H100s.
- **Adaptive importance sampling** in high dimensions; Bernstein-tight sample budgets.
- **Tighter integration** with safe-RL / policy-certification pipelines.

The scaling path (`imp`/ar-metaflow, W&B, S3 checkpointing) is scoped in `AMFS/infra/lambda_labs_notes.md`.

References — 1/3

1. L. Molu, "LevelSetPy: A GPU-Accelerated Package for Hyperbolic Hamilton-Jacobi PDEs," *ACM Trans. Math. Softw.* 51(2), Art. 10, 2025.
2. L. Molu, "The Python LevelSet Toolbox (LevelSetPy)," *IEEE CDC*, 2024.
3. M. G. Crandall, P.-L. Lions, "Viscosity Solutions of Hamilton-Jacobi Equations," *Trans. AMS* 277(1), 1983; and the two-approximations $O(\sqrt{\delta})$ error, *Math. Comp.*, 1984.
4. R. Isaacs, *Differential Games*, Wiley, 1965.
5. I. Mitchell, A. Bayen, C. Tomlin, "A Time-Dependent HJ Formulation of Reachable Sets for Continuous Dynamic Games," *IEEE TAC* 50(7), 2005.

References — 2/3

6. J. Lygeros, "On Reachability and Minimum Cost Optimal Control," *Automatica*, 2004.
7. S. Osher, R. Fedkiw, *Level Set Methods and Dynamic Implicit Surfaces*, Springer, 2003.
8. X.-D. Liu, S. Osher, T. Chan, "Weighted Essentially Non-Oscillatory Schemes," *J. Comput. Phys.*, 1994;
C.-W. Shu, S. Osher, TVD-RK, 1988.
9. L. C. Evans, *Partial Differential Equations*, AMS (heat kernel, Feynman-Kac).
10. J. Darbon, S. Osher (2016); Y. T. Chow, J. Darbon, S. Osher, W. Yin (2017) — Hopf/Lax-Oleinik grid-free formulas.
11. H. J. Kappen (2005); E. Theodorou, J. Buchli, S. Schaal (2010) — path-integral / stochastic HJB linearization.

References — 3/3

12. S. Bansal, M. Chen, S. Herbert, C. Tomlin, "DeepReach," *ICRA*, 2021; M. Chen, S. Herbert et al., system decomposition.
13. S. Summers, J. Lygeros (2010); K. Lesser, M. Oishi, R. Erwin — sample-based stochastic reachability.
14. J. Li, A. Tinka, S. Kiesel, J. W. Durham, T. K. S. Kumar, S. Koenig, "Lifelong MAPF in Large-Scale Warehouses (RHCR)," *AAAI*, 2021.
15. K. Okumura, "LaCAM / Engineering LaCAM*," 2023; PIBT and lifelong learning-based MAPF (2024-25).
16. S. Holm, "A Simple Sequentially Rejective Multiple Test Procedure," *Scand. J. Stat.*, 1979.

 **Reference ledger complete.**

Code: github.com/robotsorcerer/levelsetpy ([monte_carlo](#), [monte_carlo/AMFS](#)) · Contact: ogunmolu@amazon.com

References — Reachability Primer

17. I. Mitchell, "Reach Sets and the Hamilton-Jacobi Equation," UBC CS542D lecture notes, 2004 (with A. Bayen, M. Oishi, C. Tomlin). cs.ubc.ca/~mitchell/Class/CS542D.2004/Handouts/reachSets1.pdf
18. L. C. Evans, P. E. Souganidis, "Differential Games and Representation Formulas for Solutions of Hamilton-Jacobi-Isaacs Equations," *Indiana Univ. Math. J.*, 1984.
19. A. W. Merz, "The Homicidal Chauffeur," *AIAA J.*, 1972; I. Mitchell, numerical validation, 2001.
20. M. Falcone (static HJ); J.-P. Aubin & P. Saint-Pierre (viability kernels) — alternative Eulerian reachability.



Thank You

Questions?



ogunmolu@amazon.com

Certified safety is the sign of a value function —
HJ-Gauss computes that sign where grids cannot.

Backup slides: full proofs (residual Duhamel bound, contraction), murmuration topology theorems, AMFS harness internals, Lambda Labs scaling plan.